simplydata dictionary

Every word, A to Z, with the page of the manual that explains it

Version 0.1.0 • 284 words • 392 names from other tools • made from the code by docs/make_dictionary.py

How to use this dictionary

Keep it open next to the manual (simplydata_manual.pdf).

1. **Look up a word.** The words are in A to Z order. The words at the top of each page are the first and the last word on that page, as in a printed dictionary.
2. **Read what it means.** One line says what the word does, in the words of sdata.help("word").
3. **Try the example.** Every example runs as written. Most use a practice table, named in brackets after the example: (marks) means table = sdata.table(sdata.example("marks")) for a table word, or chart = sdata.chart(sdata.example("marks")) for a chart word.
4. **Read the page. Manual p.** gives the page of the manual where the word is explained in full; the number is a link that opens the manual at that page. **Lesson** gives the lessons that teach it.

Some words belong to more than one thing: save saves a chart, a table or numbers. Then the word has one entry with a part for each, marked chart, table, nums and so on:

sdata	sdata.chart(...)	called on simplydata itself
table	table.keep_rows(...)	a step on a table
chart	chart.bars(...)	a step on a chart
option	show="median"	an option inside a chart step
nums	nums.average()	a step on numbers: nums = sdata.numbers([12, 15, 9])
grid	grid.row(1)	a step on a grid: grid = sdata.number_grid([[3, 5], [8, 6]])
dist	dist.chance(above=60)	a step on a distribution: dist = sdata.distribution("normal")

Know the name from pandas, R, NumPy or matplotlib (groupby, iloc, hist, curve_fit)? Look it up in **Coming from another tool?** at the end.

How the words are built

- **A word says what it does, in full English words**: keep_rows, average_of, compare_averages. No short forms to remember.
- **Words come in families.** Learn keep_rows and you can guess keep_columns, keep_numbers and keep_first_rows. The next page lists the families.
- **Steps on a table start with a verb** (keep_, remove_, add_, make_, fill_); **chart types are nouns** (bars, dots, histogram).
- **Anything for one axis ends in _x or _y**: label_y, range_x, log_scale_y. x goes across, y goes up.
- **One option name means one thing everywhere**: for_each= (a result for each group), between=[low, high], colour=, from_column=.
- **Counting starts at 1**, as people count: nums.number_at(1) is the first number, grid.row(1) the top row.
- **Every change can be taken back** with undo().

Word families

A family is learned once and used again and again: the start or the end of a word says what kind of thing it does. Each family below lists its words; each word is a link to its entry.

Family	What it does	Words
keep_...	keeps only some rows, columns or numbers; the rest go	keep_biggest_rows, keep_columns, keep_columns_by_position, keep_first_numbers, keep_first_rows, keep_last_numbers, keep_last_rows, keep_numbers, keep_numbers_by_position, keep_random_numbers, keep_random_rows, keep_rows, keep_rows_by_position, keep_smallest_rows
remove_...	takes some rows, columns or numbers away; the rest stay	remove_columns, remove_columns_by_position, remove_duplicate_numbers, remove_duplicate_rows, remove_empty_numbers, remove_extra_spaces, remove_numbers, remove_numbers_by_position, remove_rows, remove_rows_by_position, remove_rows_with_empty_cells, remove_rows_with_outliers
add_..._column	adds a new column, made from the columns you have	add_adjusted_p_value_column, add_change_column, add_column, add_date_part_column, add_group_column, add_is_empty_column, add_predicted_column, add_previous_value_column, add_rank_column, add_rolling_column, add_text_part_column
add_...	adds something else to a table: rows, groups, new axes	add_clusters, add_nmds, add_principal_components, add_rows_from
make_...	changes what a column holds: numbers, text, dates, small or capital letters	make_dates, make_lowercase, make_numbers, make_text, make_uppercase
show_... and hide_...	looks at the data, or a part of the chart, without changing anything	hide_grid, hide_legend, show_first_numbers, show_first_rows, show_grid, show_last_numbers, show_last_rows, show_legend, show_rows_with_outliers
count_...	counts rows, values, empty cells or numbers	count_bars, count_columns, count_different_values, count_each, count_empty_cells, count_empty_numbers, count_rows
..._of	sums up one column in one number (or one number for each group)	average_of, biggest_of, map_of, median_of, most_common_of, percent_of, percentile_of, principal_components_of, smallest_of, standard_deviation_of, summary_of, total_of
compare_...	a statistical test: is the difference bigger than chance? It names the test and gives the p-value	compare_average_to, compare_averages, compare_before_after, compare_counts, compare_counts_to, compare_distributions, compare_each_pair, compare_percent_to, compare_profiles, compare_shape_to, compare_spreads, compare_survival
fit_...	finds the line, curve, formula or distribution that best matches the data	fit_curve, fit_distribution, fit_formula, fit_line
find_... and solve	finds where a formula is smallest or biggest, or where it equals something	find_biggest, find_smallest, solve
..._x and ..._y	a chart word for one axis: x goes across, y goes up	flip_x, flip_y, label_x, label_y, line_at_x, line_at_y, log_scale_x, log_scale_y, order_x, order_y, range_x, range_y, rotate_labels_x, rotate_labels_y, second_y, shade_between_x, shade_between_y, tick_format_x, tick_format_y, ticks_x, ticks_y
legend_...	where the legend goes	legend_bottom, legend_left, legend_right, legend_top
line_... and shade_...	draws a line or a shaded band across the chart	line_at_x, line_at_y, line_with_slope, shade_between_x, shade_between_y
as_...	hands the data on to another library	as_numpy, as_pandas, as_polars
undo, steps, start_again	every change can be taken back	start_again, steps, undo

The words, A to Z

A

absolute_value

nums `nums.absolute_value()`

Every number without its minus sign: -3 becomes 3.

e.g. `nums.absolute_value()`

Options none

See also `square_root`, `round_to`, `power`, `log`, `divide_by`

Manual p. 219

add

nums `nums.add()`

Add a number to every number (or add numbers one by one).

e.g. `nums.add(5)`

Options `number=`

See also `subtract`, `times`, `divide_by`, `power`, `square_root`

Manual p. 217; lesson 13 (p. 72)

add_adjusted_p_value_column

table `table.add_adjusted_p_value_column()`

Adjust p-values for testing many things at once, in a new column.

e.g. `table.add_adjusted_p_value_column("q", method="benjamini-hochberg")` (marks)

Options `name=`, `from_column=`, `method=`

Other tools `fdr`, `multipletests`, `p_adjust`

See also `compare_shape_to`, `compare_spreads`, `compare_counts`, `compare_counts_to`, `compare_average_to`

Manual p. 186; lesson 14 (p. 76)

add_change_column

table `table.add_change_column()`

Add a column with the change from the row before: this value minus the previous one.

e.g. `table.add_change_column("growth", from_column="population", in_percent=True, for_each="country", sorted_by="year")` (growth)

Options `name=`, `from_column=`, `in_percent=`, `rows_back=`, `for_each=`, `sorted_by=`

Other tools `diff`, `pct_change`

See also `add_previous_value_column`, `add_rolling_column`

Manual p. 175; lesson 11 (p. 57)

add_clusters

table `table.add_clusters()`

Sort the rows into groups of similar rows (clusters), in a new column.

e.g. `table.add_clusters(how_many=3)` (meadow)

Options `columns=`, `how_many=`, `method=`, `same_scale=`, `seed=`, `name=`, `linkage=`

Other tools `hclust`, `kmeans`

See also `add_principal_components`, `add_nmds`, `principal_components_of`, `distances_between_rows`, `compare_profiles`

Manual p. 203; lesson 17 (p. 90)

add_column

table `table.add_column()`

Add a new column, worked out from other columns in plain words or with a formula.

e.g. `table.add_column("study_and_marks", by_adding=["hours", "marks"])` (marks)

Options `name=`, `formula=`, `by_adding=`, `by_subtracting=`, `by_multiplying=`, `by_dividing=`, `by_joining_text=`, `separator=` and 8 more

Other tools `assign`, `mutate`, `transform`, `with_columns`

See also `add_group_column`, `add_rank_column`, `sort_by`

Manual p. 170; lesson 11 (p. 57)

add_date_part_column

table `table.add_date_part_column()`

Add a column with one part of each date, such as its year, month or weekday.

e.g. `table.add_date_part_column("weekday", from_column="date", part="weekday name")` (shop)

Options `name=`, `from_column=`, `part=`

See also `add_text_part_column`, `split_column`, `replace_text`

Manual p. 172; lesson 11 (p. 57)

add_group_column

table `table.add_group_column()`

Add a column that puts each row into a group, like turning marks into grades.

e.g. `table.add_group_column("grade", from_column="marks", cut_at=[40, 60, 80], names=["D", "C", "B", "A"])` (marks)

Options `name=`, `from_column=`, `cut_at=`, `names=`, `rules=`, `otherwise=`

Other tools `cut`, `qcut`

See also `add_column`, `add_rank_column`, `sort_by`

Manual p. 171; lesson 11 (p. 57), lesson 14 (p. 76)

add_is_empty_column

table `table.add_is_empty_column()`

Add a true/false column: True where a cell of `from_column` is empty.

e.g. `table.add_is_empty_column("sales_was_empty", from_column="sales")` (shop)

Options `name=`, `from_column=`

Other tools `is_na`, `is_null`, `isna`, `isnull`

See also `fill_empty_cells`, `remove_rows_with_empty_cells`, `count_empty_cells`

Manual p. 164; lesson 11 (p. 57)

add_nmds

table `table.add_nmds()`

Draw the rows as points on a map where close points are similar rows (NMDS), in new columns.

e.g. `table.add_nmds()` (meadow)

Options `columns=`, `measure=`, `dimensions=`, `tries=`, `seed=`

Other tools `metaMDS`, `nmds`

See also `add_clusters`, `add_principal_components`, `distances_between_rows`, `compare_profiles`, `principal_components_of`

Manual p. 204; lesson 17 (p. 90)

add_predicted_column

table `table.add_predicted_column()`

Add a column with the value the model predicts for each row.

e.g. `table.add_predicted_column("marks", using=["hours", "breakfast"])` (school)

Options `column=`, `using=`, `outcome=`, `name=`, `grouped_by=`, `start=`, `between=`, `weights=` and 1 more

See also `explain`

Manual p. 200; lesson 16 (p. 86)

add_previous_value_column

table `table.add_previous_value_column()`

Add a column with the value from the row before (or `rows_back=` rows before).

e.g. `table.add_previous_value_column("last_month", from_column="temperature", for_each="city", sorted_by="month_number")` (weather)

Options `name=`, `from_column=`, `rows_back=`, `for_each=`, `sorted_by=`

Other tools `lag`, `shift`

See also `add_rolling_column`, `add_change_column`

Manual p. 175

add_principal_components

table `table.add_principal_components()`

Add principal components: a few new columns that hold most of the spread of many columns.

e.g. `table.add_principal_components(how_many=2)` (meadow)

Options `columns=`, `how_many=`, `same_scale=`

See also `add_clusters`, `add_nmds`, `principal_components_of`, `distances_between_rows`, `compare_profiles`

Manual p. 201; lesson 17 (p. 90)

add_rank_column

table `table.add_rank_column()`

Add a column with each row's place, where 1 is the best.

e.g. `table.add_rank_column("place", from_column="marks")` (marks)

Options `name=`, `from_column=`, `biggest_first=`

Other tools `rank`

See also `add_group_column`, `add_column`, `sort_by`

Manual p. 171; lesson 11 (p. 57)

add_rolling_column

table `table.add_rolling_column()`

Add a rolling (moving) summary: for each row, the average of this row and the rows just before it.

e.g. `table.add_rolling_column("rain_3_months", from_column="rain", window=3, show="total", for_each="city", sorted_by="month_number")` (weather)

Options `name=`, `from_column=`, `window=`, `show=`, `for_each=`, `sorted_by=`

Other tools `rolling`, `rollmean`

See also `add_previous_value_column`, `add_change_column`

Manual p. 174; lesson 11 (p. 57)

add_rows_from

table `table.add_rows_from()`

Put the rows of another table under this one; columns are matched by their names.

e.g. `table.add_rows_from(sdata.example("marks"))` (marks)

Options `other=`

Other tools `append`, `bind_rows`, `concat`, `rbind`, `vstack`

See also `join_with`, `turn_columns_into_rows`, `turn_rows_into_columns`

Manual p. 209; lesson 12 (p. 64)

add_text_part_column

table `table.add_text_part_column()`

Add a column with one part of each text, such as its first letters or the text before a mark.

e.g. `table.add_text_part_column("initial", from_column="name", first_characters=1)` (marks)

Options `name=`, `from_column=`, `first_characters=`, `last_characters=`, `characters=`, `before=`, `after=`, `pattern=`

See also `add_date_part_column`, `split_column`, `replace_text`

Manual p. 172; lesson 11 (p. 57)

all_dots

chart `chart.all_dots()`

Every value as a dot, spread out so no dot hides another.

e.g. `chart.all_dots(x="class", y="marks")` (marks)

Options `x=`, `y=`, `colour=`, `transparency=`

Other tools `geom_jitter`, `stripplot`, `swarmplot`

See also `count_bars`, `side_histograms`, `density`, `qq_dots`, `smooth_line`

Manual p. 127; lesson 4 (p. 21), lesson 15 (p. 81)

animate

chart `chart.animate()`

A moving chart with one picture for each value of a column.

e.g. `chart.animate(over="year")` (growth)

Options over, seconds_per_frame=, keep=

See also zoom_box, grid, one_above_other, side_by_side, one_chart_per

Manual p. 150; lesson 8 (p. 41)

answers

sdata `sdata.answers()`

The answers to a day's warm-up questions, and a worked answer to its task.

e.g. `sdata.answers(5)`

Options number=, track=

See also day, next_day, today, start_30_days

Manual p. 240

area

chart `chart.area()`

A line with the space under it coloured in.

e.g. `chart.area(x="year", y="population", colour="country")` (growth)

Options x=, y=, colour=, stacked=, transparency=, thickness=

See also dots, histogram, lines, boxes, bars

Manual p. 115; lesson 4 (p. 21)

as_numpy

nums `nums.as_numpy()`

Give back the numbers as a plain NumPy array, for other Python tools.

e.g. `nums.as_numpy()`

Options none

See also save, chart, table, start_again, steps

Manual p. 237

as_pandas

table `table.as_pandas()`

Give back the table as a plain pandas table, for other Python tools.

e.g. `table.as_pandas()` (marks)

Options none

See also as_polars, chart, save, start_again, steps

Manual p. 212; lesson 15 (p. 81)

as_polars

table `table.as_polars()`

Give back the table as a plain Polars table, for other Python tools.

e.g. `table.as_polars()` (marks)

Options none

See also as_pandas, chart, save, start_again, steps

Manual p. 213

average

nums `nums.average()`

The average (mean) of the numbers; empty values are skipped.

e.g. `nums.average()`

Options none

See also median, total, smallest, biggest, standard_deviation

Manual p. 226; lesson 13 (p. 72)

dist `dist.average()`

The average (mean) of the distribution.

e.g. `dist.average()`

Options none

See also middle_range, median, cut_off, standard_deviation, percentile

Manual p. 197

average_of

table `table.average_of()`

The average (mean) of a column; empty cells are skipped.

e.g. `table.average_of("marks", for_each="class")` (marks)

Options column, for_each=, every=, across=

Other tools agg, aggregate, group_by, groupby, mean

See also count_each, median_of, count_rows, total_of, smallest_of

Manual p. 177; lesson 12 (p. 64)

B

band

chart `chart.band()`

A shaded band between a low and a high column, along x.

e.g. `chart.band(x="month", between=["lowest", "highest"])` (weather)

Options x=, between=, colour=, label=, transparency=

Other tools fill_between, geom_ribbon

See also dots_3d, empty_cells, surface, contour, polar

Manual p. 120; lesson 6 (p. 31)

bar *see* bars

bars

chart `chart.bars()`

Bars.

e.g. `chart.bars(x="name", y="marks")` (marks)

Options x=, y=, colour=, show=, error=, stacked=, sideways=, transparency=

Also written bar

Other tools barh, barplot, geom_bar, geom_col

See also lines, dots, area, histogram, boxes

Manual p. 113; lesson 1 (p. 10)

biggest

nums `nums.biggest()`

The biggest number; empty values are skipped.

e.g. `nums.biggest()`

Options none

See also `smallest`, `standard_deviation`, `total`, `percentile`, `median`

Manual p. 227

biggest_of

table `table.biggest_of()`

The biggest number in a column; empty cells are skipped.

e.g. `table.biggest_of("temperature", for_each="city")` (weather)

Options `column`, `for_each=`, `every=`, `across=`

Other tools `max`, `maximum`

See also `smallest_of`, `standard_deviation_of`, `total_of`, `summary_of`, `median_of`

Manual p. 178; lesson 12 (p. 64)

box *see* boxes

boxes

chart `chart.boxes()`

Boxes that show the middle half of the values and the median.

e.g. `chart.boxes(x="class", y="marks")` (marks)

Options `x=`, `y=`, `colour=`, `transparency=`, `thickness=`

Also written `box`

Other tools `boxplot`, `geom_boxplot`

See also `histogram`, `violins`, `area`, `heatmap`, `dots`

Manual p. 116; lesson 4 (p. 21), lesson 15 (p. 81)

C

caption_bottom

chart `chart.caption_bottom()`

A small line of text under the whole chart.

e.g. `chart.caption_bottom("Data from 2026")` (marks)

Options `text`

See also `title_top`, `label_x`, `label_y`, `note`, `numbers_on_bars`

Manual p. 132; lesson 2 (p. 13)

cell_at

table `table.cell_at()`

The value in one cell: a row counting from 1 and a column by name.

e.g. `table.cell_at(row=3, column="marks")` (marks)

Options `row=`, `column=`

Other tools `at`, `get_value`, `iat`, `item`

See also `row`, `column`, `describe_columns`, `column_names`, `show`

Manual p. 154; lesson 10 (p. 52)

chance

dist `dist.chance()`

The chance of a value in a range, as a share from 0 to 1 (printed as a percent).

e.g. `dist.chance(above=80)`

Options `value=`, `above=`, `below=`, `at_most=`, `at_least=`, `between=`

Other tools `cdf`, `pbinom`, `pnorm`, `sf`

See also `fit_distribution`, `density_at`, `compare_distributions`, `percentile`, `cut_off`

Manual p. 195; lesson 20 (p. 104)

chance_to_find

sdata `sdata.chance_to_find()`

What chance does a study have to find a real difference of a given size (its power)?

e.g. `sdata.chance_to_find(effect_size=0.5, values_per_group=30)`

Options `test=`, `effect_size=`, `difference=`, `spread=`, `values_per_group=`, `significance=`, `groups=`, `sides=`

See also `how_many_needed`

Manual p. 208; lesson 18 (p. 95)

change_column

table `table.change_column()`

Change the values of a column in place, with the same ways as `add_column()`.

e.g. `table.change_column("marks", "marks / 10")` (marks)

Options `column=`, `formula=`, `by_adding=`, `by_subtracting=`, `by_multiplying=`, `by_dividing=`, `by_joining_text=`, `separator=` and 8 more

See also `round_to`, `replace_values`, `remove_extra_spaces`, `make_uppercase`, `make_lowercase`

Manual p. 169; lesson 11 (p. 57)

chart

sdata `sdata.chart()`

Start a chart.

e.g. `sdata.chart(sdata.example("marks"))` (marks)

Options `data=`, `sheet=`, `fast=`

Other tools `figure`, `ggplot`, `plot`, `subplots`

See also `example`, `show`, `save`

Manual p. 111; lesson 1 (p. 10), lesson 15 (p. 81)

table `table.chart()`

Start a chart from this table.

e.g. `table.chart()` (marks)

Options none

Other tools `ggplot`, `plot`

See also `save`, `as_pandas`, `start_again`, `as_polars`, `steps`

Manual p. 212; lesson 12 (p. 64), lesson 15 (p. 81)

nums `nums.chart()`

Start a chart from these numbers.

e.g. `nums.chart()`

Options none

Other tools `hist`, `plot`

See also `table`, `save`, `start_again`, `as_numpy`, `steps`

Manual p. 236; lesson 13 (p. 72)

dist `dist.chart()`

A chart of the distribution: its density curve (or bars for counts).

e.g. `dist.chart()`

Options show=, between=

See also random_numbers, table, summary, parameters, variance

Manual p. 199; lesson 20 (p. 104)

color *see* colour

color_scale *see* colour_scale

colors *see* colours

colour

chart `chart.colour()`

One colour for the whole chart.

e.g. `chart.colour("blue")` (marks)

Options name

Also written color

See also colour_scale, colours, style, size, show_grid

Manual p. 140; lesson 3 (p. 16)

colour_scale

chart `chart.colour_scale()`

The colours for numbers on a heatmap, contour map, surface or dots coloured by a number column.

e.g. `chart.colour_scale("magma")` (weather)

Options names..., middle=, log=, steps=, between=

Also written color_scale

Other tools cmap, colormap, scale_fill_viridis_c

See also colour, colours, style, size, show_grid

Manual p. 141; lesson 3 (p. 16)

colours

chart `chart.colours()`

Your own colours for the groups, in order.

e.g. `chart.colours("blue", "orange", "green")` (marks)

Options names...

Also written colors

See also colour, colour_scale, style, size, show_grid

Manual p. 141; lesson 3 (p. 16)

column

table `table.column()`

One column on its own: a column of numbers becomes `sdata.numbers`, any other a plain list.

e.g. `table.column("marks")` (marks)

Options name=

See also column_names, describe_columns, row, cell_at, show

Manual p. 153

grid `grid.column()`

One column of the grid as new numbers, counting from 1: `column(1)` is the left one.

e.g. `grid.column(1)`

Options position=

See also row, numbers, number_at, keep_rows_by_position, show_last_rows

Manual p. 231

column_names

table `table.column_names()`

The names of all columns.

e.g. `table.column_names()` (marks)

Options none

Other tools colnames, columns, names

See also column, show, describe_columns, show_last_rows, show_first_rows

Manual p. 153; lesson 9 (p. 48)

compare_average_to

table `table.compare_average_to()`

Is the average of a column different from a number you choose, or could it be chance?

e.g. `table.compare_average_to("marks", 60)` (marks)

Options column=, value=, based_on=

See also compare_before_after, compare_counts, compare_each_pair, compare_shape_to, compare_averages

Manual p. 184

compare_averages

table `table.compare_averages()`

Are the averages of the groups really different, or could it be chance?

e.g. `table.compare_averages("marks", for_each="class")` (marks)

Options column=, for_each=, based_on=, test=

Other tools anova, aov, f_oneway, kruskal, kruskal_test, mannwhitneyu, t_test, ttest, ttest_ind, wilcox_test

See also compare_each_pair, compare_before_after, compare_average_to, compare_counts, compare_shape_to

Manual p. 183; lesson 14 (p. 76), lesson 15 (p. 81)

compare_before_after

table `table.compare_before_after()`

Did the values change from one column to the other, row by row, more than chance?

e.g. `table.compare_before_after("Maths", "Science")` (marks)

Options `before=`, `after=`, `based_on=`

Other tools `paired_t_test`, `ttest_rel`, `wilcoxon`

See also `compare_each_pair`, `compare_average_to`, `compare_averages`, `compare_counts`, `compare_shape_to`

Manual p. 184

compare_counts

table `table.compare_counts()`

Does how often each value turns up differ between groups, more than chance?

e.g. `table.compare_counts("result", for_each="class")` (marks)

Options `column=`, `for_each=`, `test=`

Other tools `chi2_contingency`, `chisq_test`, `crosstab`, `fisher_exact`

See also `compare_average_to`, `compare_shape_to`, `compare_before_after`, `compare_each_pair`, `compare_spreads`

Manual p. 185; lesson 14 (p. 76)

compare_counts_to

table `table.compare_counts_to()`

Does how often each value turns up match the shares you expect, more than chance?

e.g. `table.compare_counts_to("breakfast", expected={"yes": 0.6, "no": 0.4})` (school)

Options `column=`, `expected=`

See also `compare_spreads`, `compare_percent_to`, `compare_shape_to`, `compare_counts`, `compare_average_to`

Manual p. 187; lesson 14 (p. 76)

compare_distributions

table `table.compare_distributions()`

Fit several distributions to a column and rank them by AIC (smaller is better).

e.g. `table.compare_distributions("sales")` (shop)

Options `column=`, `distributions=`, `for_each=`

See also `fit_distribution`, `distribution`, `chance`, `density_at`, `percentile`

Manual p. 194; lesson 20 (p. 104)

compare_each_pair

table `table.compare_each_pair()`

Which pairs of groups differ? Every pair is compared, with Holm's correction.

e.g. `table.compare_each_pair("marks", for_each="class")` (marks)

Options `column=`, `for_each=`, `based_on=`, `test=`

See also `compare_averages`, `compare_before_after`, `compare_average_to`, `compare_counts`, `compare_shape_to`

Manual p. 183; lesson 14 (p. 76)

compare_percent_to

table `table.compare_percent_to()`

Is the percent of rows with one value different from a percent you choose, or could it be chance?

e.g. `table.compare_percent_to("passed", value="yes", percent=50)` (school)

Options `column=`, `value=`, `percent=`

See also `compare_counts_to`, `compare_spreads`, `compare_shape_to`, `compare_counts`, `compare_average_to`

Manual p. 187; lesson 14 (p. 76)

compare_profiles

table `table.compare_profiles()`

Do the groups differ in their make-up across many columns (PERMANOVA), or could it be chance?

e.g. `table.compare_profiles(for_each="grazed")` (meadow)

Options `columns=`, `for_each=`, `measure=`, `permutations=`, `seed=`

Other tools `adonis`, `adonis2`, `permanova`

See also `add_nmds`, `distances_between_rows`, `add_clusters`, `principal_components_of`, `add_principal_components`

Manual p. 205; lesson 17 (p. 90)

compare_shape_to

table `table.compare_shape_to()`

Do the values look bell-shaped (normal), or are they too far from it to be chance?

e.g. `table.compare_shape_to("marks", "normal", for_each="class")` (marks)

Options `column=`, `shape=`, `for_each=`, `seed=`, `samples=`

Other tools `anderson`, `normaltest`, `shapiro`, `shapiro_test`

See also `compare_counts`, `compare_average_to`, `compare_spreads`, `compare_before_after`, `compare_counts_to`

Manual p. 185; lesson 14 (p. 76)

compare_spreads

table `table.compare_spreads()`

Do the groups differ in how spread out their values are, more than chance?

e.g. `table.compare_spreads("marks", for_each="class")` (marks)

Options `column=`, `for_each=`, `test=`

See also `compare_counts_to`, `compare_shape_to`, `compare_percent_to`, `compare_counts`, `compare_average_to`

Manual p. 186; lesson 14 (p. 76)

compare_survival

table `table.compare_survival()`

Do the groups differ in the time until the event, or could it be chance? (log-rank test)

e.g. `table.compare_survival(time="days", event="died", for_each="treatment")` (trial)

Options `time=`, `event=`, `for_each=`

Other tools `logrank_test`, `survdiff`

See also `survival_over_time`, `explain_survival`, `survival_curves`

Manual p. 206; lesson 18 (p. 95)

contour

chart `chart.contour()`

A map where each colour band joins places with the same height.

e.g. `chart.contour(x="east", y="north", z="height")` (island)

Options `x=`, `y=`, `z=`, `filled=`, `transparency=`, `thickness=`

Other tools `contourf`

See also `polar`, `surface`, `pie`, `dots_3d`, `heatmap`

Manual p. 118; lesson 8 (p. 41)

correlation_between

table `table.correlation_between()`

How strongly columns of numbers go up or down together, from -1 to 1.

e.g. `table.correlation_between("hours", "marks")` (marks)

Options `columns...`, `method=`, `show=`

Other tools `cor`, `cor_test`, `corr`, `kendalltau`, `pearsonr`, `spearmanr`

See also `most_common_of`, `percentile_of`, `count_different_values`, `percent_of`, `summary_of`

Manual p. 182; lesson 12 (p. 64)

count

nums `nums.count()`

How many numbers there are (empty ones included).

e.g. `nums.count()`

Options none

See also `count_empty_numbers`, `number_at`, `show_last_numbers`, `show_first_numbers`, `show`

Manual p. 216

count_bars

chart `chart.count_bars()`

Bars that count how many rows each group has.

e.g. `chart.count_bars(x="fruit", colour="shop")` (fruit)

Options `x=`, `colour=`, `sideways=`, `transparency=`

Other tools `countplot`

See also `density`, `all_dots`, `smooth_line`, `side_histograms`, `curve`

Manual p. 126

count_columns

grid `grid.count_columns()`

How many columns the grid has.

e.g. `grid.count_columns()`

Options none

See also `count_rows`, `shape`, `number_grid`, `show_first_rows`, `show_last_rows`

Manual p. 229

count_different_values

table `table.count_different_values()`

How many different values a column has, or how many in each group.

e.g. `table.count_different_values("name", for_each="class")` (marks)

Options `column=`, `for_each=`, `every=`

Other tools `n_distinct`, `n_unique`, `nunique`

See also `count_each`, `count_rows`, `percent_of`, `percentile_of`, `summary_of`

Manual p. 181; lesson 12 (p. 64)

count_each

table `table.count_each()`

Each different value in a column, how many times it appears, and its percent.

e.g. `table.count_each("fruit")` (fruit)

Options `column`

Other tools `table`, `tabulate`, `value_counts`

See also `count_rows`, `count_different_values`, `average_of`, `median_of`, `total_of`

Manual p. 176; lesson 9 (p. 48)

nums `nums.count_each()`

Each different number, how many times it appears, and its percent, as a table.

e.g. `nums.count_each()`

Options none

Other tools `tabulate`, `value_counts`

See also `percentile`, `standard_deviation`, `biggest`, `smallest`, `total`

Manual p. 228

count_empty_cells

table `table.count_empty_cells()`

Show how many empty cells each column has.

e.g. `table.count_empty_cells()` (shop)

Options none

See also `remove_rows_with_empty_cells`, `fill_empty_cells`, `add_is_empty_column`

Manual p. 163; lesson 11 (p. 57)

count_empty_numbers

nums `nums.count_empty_numbers()`

How many values are empty (missing).

e.g. `nums.count_empty_numbers()`

Options none

See also `count`, `number_at`, `show_last_numbers`, `show_first_numbers`, `show`

Manual p. 217

count_rows

table `table.count_rows()`

How many rows the table has, or how many in each group.

e.g. `table.count_rows(for_each="class")` (marks)

Options `for_each=`, `across=`, `every=`, `in_percent=`

Other tools `count`, `height`, `len`, `nrow`, `shape`

See also `count_each`, `count_different_values`, `average_of`, `median_of`, `total_of`

Manual p. 176; lesson 9 (p. 48), lesson 12 (p. 64)

grid `grid.count_rows()`

How many rows the grid has.

e.g. `grid.count_rows()`

Options none

See also `count_columns`, `number_grid`, `shape`, `show_first_rows`, `show_last_rows`

Manual p. 229

cumulative_share

chart `chart.cumulative_share()`

A line that climbs from 0 to 1: the share of values at or below each x.

e.g. `chart.cumulative_share(x="marks", colour="class")` (marks)

Options `x=`, `colour=`, `transparency=`, `thickness=`

Other tools `ecdf`, `ecdfplot`, `stat_ecdf`

See also `qq_dots`, `pairs`, `side_histograms`, `all_dots`, `count_bars`

Manual p. 128; lesson 5 (p. 26)

curve

chart `chart.curve()`

Draw a formula, or a distribution, as a curve.

e.g. `chart.curve("y = 100 * exp(-0.3 * x)", between=[0, 10])` (marks)

Options `formula_or_distribution=`, `between=`, `steps=`, `dashed=`, `colour=`, `label=`, `thickness=`, `show=`

Other tools `fplot`, `geom_function`, `stat_function`

See also `fit_distribution`, `smooth_line`, `fit_formula`, `density`, `fit_curve`

Manual p. 124; lesson 20 (p. 104)

cut_off

dist `dist.cut_off()`

The value with this chance above it (or below it): critical values.

e.g. `dist.cut_off(above=0.025)`

Options `above=`, `below=`

See also `percentile`, `middle_range`, `density_at`, `average`, `chance`

Manual p. 196; lesson 20 (p. 104)

D

day

sdata `sdata.day()`

Show any day of the 30 days, without changing where you are.

e.g. `sdata.day(5)`

Options `number=`, `track=`

See also `today`, `answers`, `start_30_days`, `next_day`

Manual p. 240

density

chart `chart.density()`

A smooth curve that is high where many values are.

e.g. `chart.density(x="marks", colour="class")` (marks)

Options `x=`, `colour=`, `transparency=`, `thickness=`

Other tools `geom_density`, `kde`, `kdeplot`

See also `smooth_line`, `count_bars`, `curve`, `all_dots`, `fit_distribution`

Manual p. 126; lesson 5 (p. 26)

density_at

dist `dist.density_at()`

The height of the density curve at a value (continuous distributions).

e.g. `dist.density_at(60)`

Options `value=`

Other tools `dbinom`, `dnorm`, `pdf`, `pmf`

See also `chance`, `percentile`, `fit_distribution`, `cut_off`, `compare_distributions`

Manual p. 196

describe_columns

table `table.describe_columns()`

Describe every column: its kind, empty cells, smallest, biggest and average.

e.g. `table.describe_columns()` (marks)

Options none

Other tools `describe`, `dtypes`, `glimpse`, `info`, `str`

See also `column_names`, `column`, `show`, `row`, `show_last_rows`

Manual p. 153; lesson 9 (p. 48), lesson 15 (p. 81)

distances_between_rows

table `table.distances_between_rows()`

How different each row is from every other row: a square table of distances.

e.g. `table.distances_between_rows(names="plot")` (meadow)

Options `columns=`, `measure=`, `names=`

Other tools `pdist`, `vegdist`

See also `add_clusters`, `add_nmds`, `principal_components_of`, `compare_profiles`, `add_principal_components`

Manual p. 203; lesson 17 (p. 90)

distribution

sdata `sdata.distribution()`

Make a distribution to ask chances, percentiles and random numbers of.

e.g. `sdata.distribution("normal", average=50, standard_deviation=10)`

Options `name=`, `values...`

Other tools `norm`, `rv_continuous`

See also `fit_distribution`, `compare_distributions`, `chance`, `density_at`, `percentile`

Manual p. 193; lesson 20 (p. 104)

divide_by

nums `nums.divide_by()`

Divide every number by a number.

e.g. `nums.divide_by(4)`

Options `number=`

See also `times`, `power`, `subtract`, `square_root`, `add`

Manual p. 218

dots

chart `chart.dots()`

Dots (a scatter chart).

e.g. `chart.dots(x="hours", y="marks")` (marks)

Options `x=`, `y=`, `colour=`, `size=`, `shape=`, `transparency=`

Also written `scatter`

Other tools `geom_point`, `scatterplot`

See also `dots_3d`, `lines`, `area`, `bars`, `histogram`

Manual p. 114; lesson 4 (p. 21), lesson 15 (p. 81), lesson 17 (p. 90)

dots_3d

chart `chart.dots_3d()`

Dots floating in a 3D box.

e.g. `chart.dots_3d(x="hours", y="height", z="marks", colour="class")` (marks)

Options `x=`, `y=`, `z=`, `colour=`, `transparency=`

See also `dots`, `surface`, `band`, `contour`, `empty_cells`

Manual p. 119

E

empty_cells

chart `chart.empty_cells()`

A map of the empty cells: one row of the chart for each row of the table, one strip for each column, with empty cells coloured.

e.g. `chart.empty_cells()` (shop)

Options `columns=`

See also `band`, `dots_3d`, `surface`, `contour`, `polar`

Manual p. 120; lesson 11 (p. 57)

error

option `error=`

Error bars on top of averages: how spread out, or how sure, the values are.

e.g. `chart.bars(x="class", y="marks", error="standard deviation")` (marks)

See also `error_bars`, `show`, `fit_line`, `fit_curve`, `fit_formula`

Manual p. 121; lesson 5 (p. 26)

error_bars

chart `chart.error_bars()`

Error bars on dots or lines.

e.g. `chart.error_bars("standard error")` (marks)

Options `kind=`, `column=`, `dodge=`

Other tools `errorbar`, `geom_errorbar`

See also `error`, `fit_line`, `show`, `fit_curve`, `fit_formula`

Manual p. 122

example

sdata `sdata.example()`

A small made-up table to practise with.

e.g. `sdata.example("marks")` (marks)

Options `name=`

See also `chart`, `show`, `save`

Manual p. 111; lesson 1 (p. 10)

exp

nums `nums.exp()`

e (about 2.718) to the power of every number; it undoes `log()`.

e.g. `nums.exp()`

Options none

See also `log`, `running_total`, `round_to`, `absolute_value`, `square_root`

Manual p. 220

explain

table `table.explain()`

Explain one column from several others with a model, and say how big each effect is.

e.g. `table.explain("marks", using=["hours", "breakfast"])` (school)

Options `column=`, `using=`, `outcome=`, `grouped_by=`, `empty_cells=`, `seed=`

Other tools `glm`, `linregress`, `lm`, `lmer`, `logit`, `ols`

See also `add_predicted_column`

Manual p. 200; lesson 16 (p. 86)

explain_survival

table `table.explain_survival()`

Which columns go with a higher or lower risk of the event? (Cox model, hazard ratios)

e.g. `table.explain_survival(time="days", event="died", using=["treatment", "age"])` (trial)

Options `time=`, `event=`, `using=`

Other tools `cox`, `coxph`

See also `compare_survival`, `survival_curves`, `survival_over_time`

Manual p. 206; lesson 18 (p. 95)

F

fill_empty_cells

table `table.fill_empty_cells()`

Fill the empty cells of a column with a value, or with the average, median, value above or value below.

e.g. `table.fill_empty_cells("customers", with_the="median")` (shop)

Options `column=`, `with_value=`, `with_the=`, `sorted_by=`, `for_each=`

Other tools `approx`, `approxfun`, `CubicSpline`, `fill_nan`, `fill_null`, `fillna`, `interp`, `interp1d`, `interpolate`, `pchip`, `replace_na`, `spline`, `splinefun`

See also `remove_rows_with_empty_cells`, `add_is_empty_column`, `count_empty_cells`

Manual p. 164; lesson 11 (p. 57)

fill_empty_numbers

nums `nums.fill_empty_numbers()`

Fill the empty values with a number, or with the average, median, value above or value below.

e.g. `nums.fill_empty_numbers(with_value=0)`

Options `with_value=`, `with_the=`

See also `replace_numbers`, `sort`, `remove_numbers_by_position`, `reverse`, `keep_numbers_by_position`

Manual p. 225

find_biggest

sdata `sdata.find_biggest()`

Find where a formula is biggest, and its value there.

e.g. `sdata.find_biggest("3*x + 2*y", rules=["x + y <= 4", "x + 3*y <= 6", "x >= 0", "y >= 0"])`

Options `formula=`, `between=`, `start=`, `rules=`, `whole_numbers=`, `search=`, `method=`, `known=` and 1 more

Other tools `argmax`, `fmax`, `maximise`, `maximize`

See also `find_smallest`, `solve`

Manual p. 191; lesson 19 (p. 99)

find_smallest

sdata `sdata.find_smallest()`

Find where a formula is smallest, and its value there.

e.g. `sdata.find_smallest("x**2 - 4*x + 1", between=[0, 5])`

Options `formula=`, `between=`, `start=`, `rules=`, `whole_numbers=`, `search=`, `method=`, `known=` and 1 more

Other tools `argmin`, `fmin`, `fminbound`, `fminsearch`, `linear_programming`, `linprog`, `lp`, `milp`, `minimise`, `minimize`, `minimize_scalar`, `nlm`, `nlminb`, `optim`, `optimise`, `optimize`

See also `find_biggest`, `solve`

Manual p. 190; lesson 19 (p. 99)

fit_curve

table `table.fit_curve()`

Fit a curve y = b0 + b1 * x + b2 * x**2 ... that best follows the points.

e.g. `table.fit_curve(x="hours", y="marks", degree=2)` (marks)

Options `x=`, `y=`, `degree=`, `weights=`, `for_each=`

Other tools `polyfit`

See also `fit_line`, `fit_formula`

Manual p. 189; lesson 19 (p. 99)

chart `chart.fit_curve()`

A curve that fits the dots best.

e.g. `chart.fit_curve(degree=2)` (weather)

Options `degree=`, `show_equation=`, `colour=`, `thickness=`

Other tools `polyfit`

See also `fit_line`, `fit_formula`, `fit_distribution`, `error_bars`, `error`

Manual p. 123; lesson 5 (p. 26), lesson 19 (p. 99)

fit_distribution

table `table.fit_distribution()`

Fit a distribution to a column by maximum likelihood; gives back the fitted distribution.

e.g. `table.fit_distribution("sales", "lognormal")` (shop)

Options `column=`, `distribution=`, `for_each=`, `fixed=`

Other tools `fit`, `fit_distribution`, `fitdist`, `fitdistr`

See also `distribution`, `compare_distributions`, `chance`, `density_at`, `percentile`

Manual p. 194; lesson 20 (p. 104)

chart `chart.fit_distribution()`

The best-fitting distribution's curve over a histogram or density chart.

e.g. `chart.fit_distribution("normal")` (marks)

Options `distribution=`, `colour=`, `thickness=`

Other tools `fitdist`, `fitdistr`

See also `fit_formula`, `fit_curve`, `fit_line`, `curve`, `smooth_line`

Manual p. 124; lesson 20 (p. 104)

nums `nums.fit_distribution()`

Fit a distribution to the numbers by maximum likelihood; gives back the fitted distribution.

e.g. `nums.fit_distribution("gamma")`

Options `distribution=`, `fixed=`

Other tools `fit`, `fitdist`, `fitdistr`

See also compare_distributions, chance, density_at, distribution, percentile

Manual p. 195

fit_formula

table table.fit_formula()

Find the values in a formula that make it follow the data best.

e.g. table.fit_formula("population = a * exp(b * (year - 2000))") (growth)

Options formula=, start=, between=, weights=, for_each=, search=, method=

Other tools curve_fit, least_squares, leastsq, nls, nlsLM

See also fit_curve, fit_line

Manual p. 189; lesson 14 (p. 76), lesson 19 (p. 99)

chart chart.fit_formula()

The curve of a formula fitted to the dots.

e.g. chart.fit_formula("population = a * exp(b * (year - 2000))") (growth)

Options formula=, start=, between=, show_equation=, colour=, thickness=

Other tools curve_fit, nls

See also fit_curve, fit_distribution, fit_line, curve, error_bars

Manual p. 123; lesson 19 (p. 99)

fit_line

table table.fit_line()

Fit the straight line y = slope * x + intercept that best follows the points.

e.g. table.fit_line(x="hours", y="marks") (marks)

Options x=, y=

Other tools linregress, lm, ols

See also fit_curve, fit_formula

Manual p. 188; lesson 14 (p. 76), lesson 19 (p. 99)

chart chart.fit_line()

A straight line that fits the dots best (linear regression).

e.g. chart.fit_line() (marks)

Options confidence_band=, show_equation=, colour=, thickness=

Other tools geom_smooth, lmplot, regplot

See also fit_curve, fit_formula, fit_distribution, error_bars, error

Manual p. 122; lesson 5 (p. 26), lesson 7 (p. 36), lesson 15 (p. 81), lesson 19 (p. 99)

flip_x

chart chart.flip_x()

Turn the x-axis around, so numbers grow from right to left.

e.g. chart.flip_x() (marks)

Options none

See also flip_y, second_y, tick_format_y, tick_format_x, rotate_labels_y

Manual p. 140

flip_y

chart chart.flip_y()

Turn the y-axis upside down, for example for depth going down.

e.g. chart.flip_y() (island)

Options none

See also flip_x, second_y, tick_format_y, tick_format_x, rotate_labels_y

Manual p. 140; lesson 6 (p. 31)

font_name

chart chart.font_name()

The font for all the text on the chart.

e.g. chart.font_name("DejaVu Serif") (marks)

Options name

See also font_size, names_on_dots, numbers_on_bars, note, label_y

Manual p. 135

font_size

chart chart.font_size()

Make all the text bigger or smaller.

e.g. chart.font_size(14) (marks)

Options size

See also font_name, names_on_dots, numbers_on_bars, note, label_y

Manual p. 134; lesson 2 (p. 13)

G

grid

sdata sdata.grid()

Charts in rows and columns.

e.g. sdata.grid(a, b, c, d, per_row=2) (marks)

Options charts..., per_row=, same_x=, same_y=, letters=

See also one_above_other, zoom_box, side_by_side, animate, one_chart_per

Manual p. 149; lesson 7 (p. 36), lesson 15 (p. 81)

H

heatmap

chart chart.heatmap()

A grid of coloured squares, one colour for each number.

e.g. chart.heatmap(x="month", y="city", value="rain", numbers=True) (weather)

Options x=, y=, value=, numbers=, cluster=, transparency=

Other tools geom_tile, imshow, pcolormesh

See also violins, pie, boxes, polar, histogram

Manual p. 117; lesson 8 (p. 41)

help

sdata `sdata.help()`

Explain one word: what it does, its options and an example you can run.

e.g. `sdata.help("bars")`

Options `word=`

See also `words`, `lesson`

Manual p. 238; lesson 8 (p. 41)

hide_grid

chart `chart.hide_grid()`

No grid lines.

e.g. `chart.hide_grid()` (marks)

Options none

See also `hide_legend`, `show_grid`, `legend_top`, `size`, `legend_bottom`

Manual p. 143

hide_legend

chart `chart.hide_legend()`

No legend.

e.g. `chart.hide_legend()` (weather)

Options none

See also `hide_grid`, `legend_right`, `show_legend`, `legend_left`, `label_legend`

Manual p. 145; lesson 3 (p. 16)

histogram

chart `chart.histogram()`

Bars that count how many values fall in each range.

e.g. `chart.histogram(x="marks", bins=10)` (marks)

Options `x=`, `bins=`, `colour=`, `show=`, `stacked=`, `steps=`, `cumulative=`, `bin_width=` and 2 more

Other tools `geom_histogram`, `hist`, `histplot`

See also `area`, `boxes`, `dots`, `violins`, `lines`

Manual p. 115; lesson 4 (p. 21), lesson 7 (p. 36)

how_many_needed

sdata `sdata.how_many_needed()`

How many values (in each group) does a study need to find a real difference?

e.g. `sdata.how_many_needed(effect_size=0.5)`

Options `test=`, `effect_size=`, `difference=`, `spread=`, `power=`, `significance=`, `groups=`, `sides=`

Other tools `power_t_test`, `pwr`, `solve_power`

See also `chance_to_find`

Manual p. 207; lesson 18 (p. 95)

J

join_with

table `table.join_with()`

Bring in the columns of another table, matching rows on a column such as a name.

e.g. `table.join_with({"name": ["Asha", "Ben"], "age": [12, 13]}, using="name")` (marks)

Options `other=`, `using=`, `keep=`

Other tools `inner_join`, `join`, `left_join`, `merge`

See also `add_rows_from`, `turn_columns_into_rows`, `turn_rows_into_columns`

Manual p. 209; lesson 12 (p. 64)

K

keep_biggest_rows

table `table.keep_biggest_rows()`

Keep the n rows with the biggest values in a column (also for each group).

e.g. `table.keep_biggest_rows(3, by="marks", for_each="subject")` (marks)

Options `n=`, `by=`, `for_each=`

Other tools `nlargest`, `slice_max`, `top_n`

See also `keep_smallest_rows`, `keep_random_rows`, `keep_rows_by_position`, `keep_last_rows`, `keep_first_rows`

Manual p. 161; lesson 10 (p. 52)

keep_columns

table `table.keep_columns()`

Keep only these columns: by name (in the order you list them), a run of names, or by kind.

e.g. `table.keep_columns("name", "marks")` (marks)

Options `names...`, `to=`, `kind=`, `starts_with=`, `ends_with=`, `contains=`, `pattern=`

Other tools `select`, `select_dtypes`

See also `keep_columns_by_position`, `remove_columns`, `remove_columns_by_position`, `rename_column`

Manual p. 155; lesson 10 (p. 52)

keep_columns_by_position

table `table.keep_columns_by_position()`

Keep columns by their position, counting from 1 at the left; it says which columns they were.

e.g. `table.keep_columns_by_position(1, to=3)` (marks)

Options `positions=`, `to=`, `step=`

See also `keep_columns`, `remove_columns`, `remove_columns_by_position`, `rename_column`

Manual p. 156; lesson 10 (p. 52)

grid `grid.keep_columns_by_position()`

Keep columns of the grid by their position, counting from 1 at the left.

e.g. `grid.keep_columns_by_position(2, to=3)`

Options `positions=`, `to=`, `step=`

See also `keep_rows_by_position`, `remove_rows_by_position`, `numbers`, `remove_columns_by_position`, `column`

Manual p. 232

keep_first_numbers

nums `nums.keep_first_numbers()`

Keep only the first numbers.

e.g. `nums.keep_first_numbers(10)`

Options `how_many=`

See also `keep_last_numbers`, `keep_numbers`, `keep_random_numbers`, `keep_numbers_by_position`, `remove_numbers`

Manual p. 222

keep_first_rows

table `table.keep_first_rows()`

Keep only the first rows.

e.g. `table.keep_first_rows(10)` (marks)

Options `n=`

See also `keep_last_rows`, `keep_rows`, `keep_rows_by_position`, `keep_random_rows`, `keep_biggest_rows`

Manual p. 158

keep_last_numbers

nums `nums.keep_last_numbers()`

Keep only the last numbers.

e.g. `nums.keep_last_numbers(10)`

Options `how_many=`

See also `keep_first_numbers`, `keep_random_numbers`, `keep_numbers`, `keep_numbers_by_position`, `remove_numbers`

Manual p. 222

keep_last_rows

table `table.keep_last_rows()`

Keep only the last rows.

e.g. `table.keep_last_rows(10)` (marks)

Options `n=`

See also `keep_first_rows`, `keep_rows_by_position`, `keep_rows`, `keep_random_rows`, `keep_biggest_rows`

Manual p. 159

keep_numbers

nums `nums.keep_numbers()`

Keep only the numbers that match one condition.

e.g. `nums.keep_numbers(above=50)`

Options `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=`, `not_equal_to=`, `between=`, `one_of=` and 1 more

See also `keep_first_numbers`, `keep_last_numbers`, `keep_random_numbers`, `keep_numbers_by_position`, `remove_numbers`

Manual p. 221; lesson 13 (p. 72)

keep_numbers_by_position

nums `nums.keep_numbers_by_position()`

Keep numbers by their position, counting from 1: a run, every n-th number, or a list.

e.g. `nums.keep_numbers_by_position(2, to=4)`

Options `positions=`, `to=`, `step=`

See also `keep_random_numbers`, `keep_last_numbers`, `keep_first_numbers`, `keep_numbers`, `remove_empty_numbers`

Manual p. 223

keep_random_numbers

nums `nums.keep_random_numbers()`

Keep some numbers picked at random (they keep their order); a seed gives the same pick every time.

e.g. `nums.keep_random_numbers(10, seed=1)`

Options `how_many=`, `seed=`

Other tools `choice`, `sample`

See also `keep_last_numbers`, `keep_first_numbers`, `keep_numbers_by_position`, `keep_numbers`, `remove_duplicate_numbers`

Manual p. 222

keep_random_rows

table `table.keep_random_rows()`

Keep some rows picked at random; give a seed to get the same rows every time.

e.g. `table.keep_random_rows(10, seed=1)` (marks)

Options `n=`, `seed=`

Other tools `sample`, `sample_n`, `slice_sample`

See also `keep_biggest_rows`, `keep_rows_by_position`, `keep_smallest_rows`, `keep_last_rows`, `keep_first_rows`

Manual p. 160; lesson 10 (p. 52)

keep_rows

table `table.keep_rows()`

Keep only the rows that match a condition.

e.g. `table.keep_rows("marks", above=50)` (marks)

Options `column=`, `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=`, `not_equal_to=`, `between=` and 7 more

Other tools `filter`, `loc`, `query`, `subset`, `where`

See also `keep_first_rows`, `keep_last_rows`, `keep_rows_by_position`, `keep_random_rows`, `keep_biggest_rows`

Manual p. 157; lesson 10 (p. 52)

keep_rows_by_position

table `table.keep_rows_by_position()`

Keep rows by their position, counting from 1: a run, every n-th row, or a list.

e.g. `table.keep_rows_by_position(10, to=20)` (marks)

Options `positions=`, `to=`, `step=`

Other tools `gather_every`, `iloc`, `nth`, `slice`, `take`

See also `keep_last_rows`, `keep_first_rows`, `keep_random_rows`, `keep_rows`, `keep_biggest_rows`

Manual p. 159; lesson 10 (p. 52)

grid `grid.keep_rows_by_position()`

Keep rows of the grid by their position, counting from 1: a run, every n-th row, or a list.

e.g. `grid.keep_rows_by_position(1, to=2)`

Options `positions=`, `to=`, `step=`

See also `keep_columns_by_position`, `numbers`, `column`, `remove_rows_by_position`, `row`

Manual p. 232

keep_smallest_rows

table `table.keep_smallest_rows()`

Keep the n rows with the smallest values in a column (also for each group).

e.g. `table.keep_smallest_rows(3, by="marks", for_each="subject")` (marks)

Options `n=`, `by=`, `for_each=`

Other tools `nsmallest`, `slice_min`

See also `keep_biggest_rows`, `keep_random_rows`, `keep_rows_by_position`, `keep_last_rows`, `keep_first_rows`

Manual p. 162

L

label_color_bar *see* `label_colour_bar`

label_colour_bar

chart `chart.label_colour_bar()`

The name beside the colour bar of a heatmap or contour.

e.g. `chart.label_colour_bar("Height (m)")` (island)

Options `text`

Also written `label_color_bar`

See also `label_legend`, `show_legend`, `hide_legend`, `legend_right`, `legend_left`

Manual p. 146; lesson 8 (p. 41)

label_legend

chart `chart.label_legend()`

The name above the legend.

e.g. `chart.label_legend("Class")` (marks)

Options `text`

See also `label_colour_bar`, `show_legend`, `hide_legend`, `legend_right`, `legend_left`

Manual p. 145; lesson 3 (p. 16)

label_x

chart `chart.label_x()`

The name under the x-axis.

e.g. `chart.label_x("Student")` (marks)

Options `text`

Other tools `set_xlabel`, `xlabel`

See also `label_y`, `caption_bottom`, `title_top`, `note`, `numbers_on_bars`

Manual p. 132; lesson 2 (p. 13)

label_y

chart `chart.label_y()`

The name beside the y-axis.

e.g. `chart.label_y("Marks")` (marks)

Options `text`

Other tools `set_ylabel`, `ylabel`

See also `label_x`, `note`, `caption_bottom`, `numbers_on_bars`, `title_top`

Manual p. 132; lesson 2 (p. 13)

legend_bottom

chart `chart.legend_bottom()`

The legend below the chart, in one row under the x-axis name.

e.g. `chart.legend_bottom()` (weather)

Options none

See also `legend_top`, `legend_left`, `legend_right`, `hide_grid`, `show_grid`

Manual p. 144; lesson 3 (p. 16)

legend_left

chart `chart.legend_left()`

The legend to the left of the chart.

e.g. `chart.legend_left()` (weather)

Options none

See also `legend_bottom`, `legend_right`, `legend_top`, `hide_legend`, `hide_grid`

Manual p. 144

legend_right

chart `chart.legend_right()`

The legend to the right of the chart, outside the plot.

e.g. `chart.legend_right()` (weather)

Options none

Other tools `legend`

See also `legend_left`, `legend_bottom`, `legend_top`, `hide_legend`, `show_legend`

Manual p. 144

legend_top

chart `chart.legend_top()`

The legend above the chart, in one row.

e.g. `chart.legend_top()` (weather)

Options none

See also `legend_bottom`, `legend_left`, `legend_right`, `hide_grid`, `show_grid`

Manual p. 143; lesson 3 (p. 16)

lesson

sdata `sdata.lesson()`

A lesson: an example to run, what each line does, and a task.

e.g. `sdata.lesson(1)`

Options `number=`, `solutions=`

See also `help`, `words`

Manual p. 239

line *see* `lines`

line_at_x

chart `chart.line_at_x()`

A straight up-and-down line at one x value.

e.g. `chart.line_at_x(2010, label="2010")` (growth)

Options `value`, `dashed=`, `colour=`, `label=`, `thickness=`

Other tools `axvline`, `geom_vline`, `vline`

See also `line_at_y`, `line_with_slope`, `shade_between_x`, `shade_between_y`

Manual p. 129; lesson 6 (p. 31)

line_at_y

chart `chart.line_at_y()`

A straight side-to-side line at one y value.

e.g. `chart.line_at_y(50, label="pass mark")` (marks)

Options `value`, `dashed=`, `colour=`, `label=`, `thickness=`

Other tools `axhline`, `geom_hline`, `hline`

See also `line_at_x`, `line_with_slope`, `shade_between_x`, `shade_between_y`

Manual p. 130; lesson 6 (p. 31)

line_with_slope

chart `chart.line_with_slope()`

A straight line with a given slope, right across the chart.

e.g. `chart.line_with_slope(1, through=(0, 0))` (marks)

Options `slope`, `through=`, `intercept=`, `dashed=`, `colour=`, `label=`, `thickness=`

Other tools `abline`, `axline`

See also `line_at_y`, `line_at_x`, `shade_between_x`, `shade_between_y`

Manual p. 130; lesson 6 (p. 31)

lines

chart `chart.lines()`

Lines.

e.g. `chart.lines(x="month", y="temperature", colour="city")` (weather)

Options `x=`, `y=`, `colour=`, `show=`, `error=`, `dots=`, `dashed=`, `thickness=` and 2 more

Also written `line`

Other tools `figure`, `geom_line`, `ggplot`, `plot`, `subplots`

See also `bars`, `dots`, `area`, `histogram`, `boxes`

Manual p. 113; lesson 3 (p. 16), lesson 4 (p. 21)

log

nums `nums.log()`

The logarithm of every number: natural (base e) unless you give `base=10` or `base=2`.

e.g. `nums.log()`

Options `base=`

See also `round_to`, `exp`, `absolute_value`, `running_total`, `square_root`

Manual p. 220

log_scale_x

chart `chart.log_scale_x()`

A log scale on the x-axis: 1, 10, 100, 1000 are evenly spaced.

e.g. `chart.log_scale_x()` (growth)

Options none

Other tools `semilogx`, `xscale`

See also `log_scale_y`, `range_y`, `range_x`, `ticks_x`, `ticks_y`

Manual p. 136

log_scale_y

chart `chart.log_scale_y()`

A log scale on the y-axis: 1, 10, 100, 1000 are evenly spaced.

e.g. `chart.log_scale_y()` (growth)

Options none

Other tools `semilogy`, `yscale`

See also `log_scale_x`, `ticks_x`, `range_y`, `ticks_y`, `range_x`

Manual p. 136; lesson 6 (p. 31)

M

make_dates

table `table.make_dates()`

Turn a column into dates; cells that are not dates become empty.

e.g. `table.make_dates("year")` (growth)

Options `column=`, `order=`

Other tools `as_date`, `to_datetime`

See also `make_text`, `make_lowercase`, `make_numbers`, `make_uppercase`, `remove_extra_spaces`

Manual p. 166; lesson 11 (p. 57)

make_lowercase

table `table.make_lowercase()`

Write all the text in a column in small letters.

e.g. `table.make_lowercase("name")` (marks)

Options `column=`

Other tools `lower`, `tolower`

See also `make_dates`, `make_uppercase`, `make_text`, `make_numbers`, `remove_extra_spaces`

Manual p. 166

make_numbers

table `table.make_numbers()`

Turn a column into numbers; cells that are not numbers become empty.

e.g. `table.make_numbers("marks")` (marks)

Options `column=`

Other tools `as_numeric`, `astype`, `cast`, `to_numeric`

See also `make_text`, `make_dates`, `make_lowercase`, `make_uppercase`, `remove_extra_spaces`

Manual p. 165; lesson 11 (p. 57)

make_text

table `table.make_text()`

Turn a column into text.

e.g. `table.make_text("month_number")` (weather)

Options `column=`

See also `make_numbers`, `make_dates`, `make_lowercase`, `make_uppercase`, `remove_extra_spaces`

Manual p. 165

make_uppercase

table `table.make_uppercase()`

Write all the text in a column in capital letters.

e.g. `table.make_uppercase("name")` (marks)

Options `column=`

Other tools `toupper`, `upper`

See also `make_lowercase`, `make_dates`, `make_text`, `make_numbers`, `remove_extra_spaces`

Manual p. 166

map_of

table `table.map_of()`

Estimate a column at evenly spaced places across one or more columns (a map), ready for a contour or heatmap.

e.g. `table.map_of("height", across=["east", "north"], steps=100)` (island)

Options `column=`, `across=`, `steps=`, `joined_by=`, `smoothness=`

Other tools `griddata`, `interp2d`, `RBFInterpolator`

See also `values_at`

Manual p. 193; lesson 20 (p. 104)

median

nums `nums.median()`

The middle number once the numbers are sorted; empty values are skipped.

e.g. `nums.median()`

Options none

See also `average`, `total`, `smallest`, `biggest`, `standard_deviation`

Manual p. 226; lesson 13 (p. 72)

dist `dist.median()`

The median: half of the distribution is below it.

e.g. `dist.median()`

Options none

See also `average`, `standard_deviation`, `middle_range`, `variance`, `cut_off`

Manual p. 197

median_of

table `table.median_of()`

The middle number of a column once it is sorted; empty cells are skipped.

e.g. `table.median_of("marks", for_each="class")` (marks)

Options `column`, `for_each=`, `every=`, `across=`

Other tools `median`

See also `average_of`, `total_of`, `count_each`, `smallest_of`, `count_rows`

Manual p. 177

middle_range

dist `dist.middle_range()`

The range that holds the middle percent of the distribution, as Numbers [low, high].

e.g. `dist.middle_range(95)`

Options `percent=`

See also `cut_off`, `average`, `percentile`, `median`, `density_at`

Manual p. 197

most_common_of

table `table.most_common_of()`

The value that turns up most often (the mode), for numbers or text; empty cells are skipped.

e.g. `table.most_common_of("days_absent", for_each="school")` (school)

Options `column=`, `for_each=`, `every=`

Other tools `mode`

See also `percentile_of`, `correlation_between`, `count_different_values`, `percent_of`, `summary_of`

Manual p. 181; lesson 12 (p. 64)

N

names_on_dots

chart `chart.names_on_dots()`

Write a name next to each dot.

e.g. `chart.names_on_dots("name")` (marks)

Options `column`, `colour=`

See also `numbers_on_bars`, `font_size`, `note`, `font_name`, `label_y`

Manual p. 134; lesson 2 (p. 13)

next_day

sdata `sdata.next_day()`

Go on to the next day now, without waiting for tomorrow.

e.g. `sdata.next_day()`

Options none

See also `answers`, `day`, `today`, `start_30_days`

Manual p. 241

note

chart `chart.note()`

A note on the chart at a place x, y.

e.g. `chart.note("Biggest", x=2024, y=49)` (growth)

Options `text`, `x=`, `y=`, `arrow=`

Other tools `annotate`, `text`

See also `label_y`, `numbers_on_bars`, `label_x`, `names_on_dots`, `caption_bottom`

Manual p. 133; lesson 8 (p. 41)

number_at

nums `nums.number_at()`

The number at a position, counting from 1: `number_at(1)` is the first number.

e.g. `nums.number_at(1)`

Options `position=`

See also `show_last_numbers`, `count`, `show_first_numbers`, `count_empty_numbers`, `show`

Manual p. 216

grid `grid.number_at()`

The number at a row and column, counting from 1.

e.g. `grid.number_at(row=1, column=2)`

Options `row=`, `column=`

See also `number_grid`, `show_last_rows`, `row`, `show_first_rows`, `column`

Manual p. 231; lesson 13 (p. 72)

number_grid

sdata `sdata.number_grid()`

Make a grid of numbers in rows and columns.

e.g. `sdata.number_grid([[1, 2, 3], [4, 5, 6]])`

Options `values=`, `rows=`, `columns=`, `filled_with=`

Other tools `full`, `matrix`, `ones`, `zeros`

See also `number_at`, `count_rows`, `count_columns`, `shape`, `show_first_rows`

Manual p. 228; lesson 13 (p. 72)

nums `nums.number_grid()`

Make a NEW grid from these numbers, filled row by row.

e.g. `nums.number_grid(columns=3)`

Options `rows=`, `columns=`

See also `number_at`, `count_rows`, `count_columns`, `shape`, `show_first_rows`

Manual p. 229

numbers

sdata `sdata.numbers()`

Make a list of numbers.

e.g. `sdata.numbers([3, 5, 2, 8])`

Options `values...`, `to=`, `step=`, `how_many=`

Other tools `arange`, `array`, `linspace`, `seq`, `seq_len`

See also `random_numbers`, `show`, `show_first_numbers`, `show_last_numbers`, `number_at`

Manual p. 214; lesson 13 (p. 72)

grid `grid.numbers()`

All the numbers of the grid as a list, row by row (flatten).

e.g. `grid.numbers()`

Options none

See also `column`, `keep_rows_by_position`, `row`, `keep_columns_by_position`, `number_at`

Manual p. 232

numbers_on_bars

chart `chart.numbers_on_bars()`

Write the value of each bar on it.

e.g. `chart.numbers_on_bars()` (fruit)

Options `decimals=`

See also `note`, `names_on_dots`, `label_y`, `font_size`, `label_x`

Manual p. 133; lesson 5 (p. 26)

O

one_above_other

sdata `sdata.one_above_other()`

Charts on top of each other.

e.g. `sdata.one_above_other(a, b)` (weather)

Options `charts...`, `same_x=`, `same_y=`, `letters=`

See also `one_chart_per`, `side_by_side`, `grid`, `zoom_box`, `animate`

Manual p. 149; lesson 7 (p. 36)

one_chart_per

chart `chart.one_chart_per()`

One small chart for each group.

e.g. `chart.one_chart_per("class", per_row=3)` (marks)

Options `by=`, `down=`, `across=`, `per_row=`, `same_x=`, `same_y=`

Other tools `facet_grid`, `facet_wrap`, `FacetGrid`, `subplots`

See also `one_above_other`, `side_by_side`, `grid`, `zoom_box`, `animate`

Manual p. 148; lesson 7 (p. 36)

order_legend

chart `chart.order_legend()`

Put the colour groups in your order: in the legend, the colours and side by side.

e.g. `chart.order_legend("Science", "Maths", "English")` (marks)

Options `names...`

See also `order_y`, `order_x`

Manual p. 147; lesson 3 (p. 16)

order_x

chart `chart.order_x()`

Put the groups along the x-axis in your order.

e.g. `chart.order_x("C", "B", "A")` (marks)

Options `names...`, `biggest_first=`

See also `order_y`, `order_legend`

Manual p. 146; lesson 3 (p. 16)

order_y

chart `chart.order_y()`

Put the groups along the y-axis in your order, the first at the top.

e.g. `chart.order_y("C", "B", "A")` (marks)

Options `names...`, `biggest_first=`

See also `order_x`, `order_legend`

Manual p. 147

P

pairs

sdata `sdata.pairs()`

Every number column against every other.

e.g. `sdata.pairs(sdata.example("marks"), colour="class")` (marks)

Options `data`, `colour=`, `columns=`

See also `cumulative_share`, `qq_dots`, `side_histograms`, `all_dots`, `count_bars`

Manual p. 129; lesson 7 (p. 36)

parameters

dist `dist.parameters()`

The parameters as a dictionary of plain names: {"average": 50, "standard_deviation": 10}.

e.g. `dist.parameters()`

Options none

See also `table`, `chart`, `random_numbers`, `summary`, `variance`

Manual p. 199

percent_of

table `table.percent_of()`

What percent of the rows meet a condition, or each group's share.

e.g. `table.percent_of("marks", above=60, for_each="class")` (marks)

Options `column=`, `for_each=`, `every=`, `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=` and 7 more

See also `summary_of`, `count_different_values`, `standard_deviation_of`, `percentile_of`, `biggest_of`

Manual p. 180; lesson 12 (p. 64)

percentile

nums `nums.percentile()`

The number that this percent of the numbers are below: `percentile(50)` is the median.

e.g. `nums.percentile(90)`

Options `percent=`

Other tools `percentile`, `quantile`

See also `standard_deviation`, `count_each`, `biggest`, `smallest`, `total`

Manual p. 228; lesson 13 (p. 72)

dist `dist.percentile()`

The value that this percent of the distribution is below: `percentile(95)`.

e.g. `dist.percentile(95)`

Options `percent=`

Other tools `isf`, `percentile`, `ppf`, `qnorm`, `quantile`

See also `density_at`, `cut_off`, `chance`, `middle_range`, `fit_distribution`

Manual p. 196; lesson 20 (p. 104)

percentile_of

table `table.percentile_of()`

The value below which a given percent of the numbers lie; empty cells are skipped.

e.g. `table.percentile_of("marks", 90, for_each="class")` (marks)

Options `column=`, `percent=`, `for_each=`, `every=`, `across=`

Other tools `percentile`, `quantile`

See also `count_different_values`, `most_common_of`, `percent_of`, `correlation_between`, `summary_of`

Manual p. 181; lesson 12 (p. 64)

pie

chart `chart.pie()`

A circle cut into slices, one slice per group.

e.g. `chart.pie(labels="fruit", values="count")` (fruit)

Options `labels=`, `values=`, `transparency=`

See also `heatmap`, `polar`, `violins`, `contour`, `boxes`

Manual p. 117; lesson 4 (p. 21)

polar

chart `chart.polar()`

Dots on a round chart: how far round, and how far out.

e.g. `chart.polar(angle="wind_direction", distance="wind_speed", colour="city")` (weather)

Options `angle=`, `distance=`, `colour=`, `degrees=`, `transparency=`

See also `pie`, `contour`, `heatmap`, `surface`, `violins`

Manual p. 118

power

nums `nums.power()`

Raise every number to a power: `power(2)` squares, `power(3)` cubes.

e.g. `nums.power(2)`

Options `number=`

See also `divide_by`, `square_root`, `times`, `absolute_value`, `subtract`

Manual p. 218

principal_components_of

table `table.principal_components_of()`

A table of the principal components: how much spread each explains and the weight of each column.

e.g. `table.principal_components_of()` (meadow)

Options `columns=`, `same_scale=`

Other tools `pca`, `prcomp`

See also `add_principal_components`, `add_clusters`, `distances_between_rows`, `add_nmds`, `compare_profiles`

Manual p. 202; lesson 17 (p. 90)

Q

qq_dots

chart `chart.qq_dots()`

Dots that lie on a straight line when the values follow a normal (bell) curve.

e.g. `chart.qq_dots("marks")` (marks)

Options `column=`, `for_each=`, `transparency=`, `thickness=`, `distribution=`

Other tools `probplot`, `qqnorm`, `qqplot`

See also `side_histograms`, `cumulative_share`, `all_dots`, `pairs`, `count_bars`

Manual p. 127; lesson 5 (p. 26)

R

random_numbers

sdata `sdata.random_numbers()`

Make random numbers; give `seed=` to get the same numbers every time.

e.g. `sdata.random_numbers(how_many=10, between=[0, 1], seed=1)`

Options `how_many=`, `between=`, `average=`, `standard_deviation=`, `whole_numbers=`, `seed=`, `rows=`, `columns=`

Other tools `rand`, `randn`, `random`, `rnorm`, `runif`, `rvs`

See also `numbers`, `show`, `show_first_numbers`, `show_last_numbers`, `number_at`

Manual p. 214; lesson 13 (p. 72)

dist `dist.random_numbers()`

Numbers drawn at random from the distribution; the same seed gives the same numbers.

e.g. `dist.random_numbers(how_many=100, seed=1)`

Options `how_many=`, `seed=`, `rows=`, `columns=`

Other tools `rand`, `randn`, `random`, `rnorm`, `runif`, `rvs`

See also `summary`, `chart`, `variance`, `table`, `standard_deviation`

Manual p. 198

range_x

chart `chart.range_x()`

From where to where the x-axis goes.

e.g. `chart.range_x(0, 10)` (marks)

Options `low=`, `high=`

Other tools `set_xlim`, `xlim`

See also `range_y`, `log_scale_x`, `log_scale_y`, `ticks_x`, `ticks_y`

Manual p. 135

range_y

chart `chart.range_y()`

From where to where the y-axis goes.

e.g. `chart.range_y(0, 100)` (marks)

Options `low=`, `high=`

Other tools `set_ylim`, `ylim`

See also `range_x`, `log_scale_x`, `log_scale_y`, `ticks_x`, `ticks_y`

Manual p. 136; lesson 6 (p. 31)

remove_columns

table `table.remove_columns()`

Remove these columns: by name, a run of names, or by kind.

e.g. `table.remove_columns("height")` (marks)

Options `names...`, `to=`, `kind=`, `starts_with=`, `ends_with=`, `contains=`, `pattern=`

Other tools `drop`

See also `remove_columns_by_position`, `keep_columns`, `keep_columns_by_position`, `rename_column`

Manual p. 155; lesson 10 (p. 52)

remove_columns_by_position

table `table.remove_columns_by_position()`

Remove columns by their position, counting from 1 at the left; it says which columns they were.

e.g. `table.remove_columns_by_position(1)` (marks)

Options `positions=`, `to=`, `step=`

See also `remove_columns`, `keep_columns_by_position`, `rename_column`, `keep_columns`

Manual p. 156

grid `grid.remove_columns_by_position()`

Remove columns of the grid by their position, counting from 1 at the left.

e.g. `grid.remove_columns_by_position(1)`

Options `positions=`, `to=`, `step=`

See also `remove_rows_by_position`, `replace_numbers`, `keep_columns_by_position`, `swap_rows_and_columns`, `keep_rows_by_position`

Manual p. 233

remove_duplicate_numbers

nums `nums.remove_duplicate_numbers()`

Remove numbers that repeat an earlier number, so each number is left once, in its first place.

e.g. `nums.remove_duplicate_numbers()`

Options none

See also `remove_empty_numbers`, `remove_numbers_by_position`, `remove_numbers`, `keep_random_numbers`, `keep_last_numbers`

Manual p. 223

remove_duplicate_rows

table `table.remove_duplicate_rows()`

Remove rows that repeat an earlier row, or that repeat it in the columns you name.

e.g. `table.remove_duplicate_rows(columns=["name"])` (marks)

Options `columns=`

Other tools `distinct`, `drop_duplicates`, `unique`

See also `remove_rows_by_position`, `remove_rows_with_outliers`, `remove_rows`, `keep_random_rows`, `keep_biggest_rows`

Manual p. 161

remove_empty_numbers

nums `nums.remove_empty_numbers()`

Remove the empty (missing) values.

e.g. `nums.remove_empty_numbers()`

Options none

See also `remove_duplicate_numbers`, `remove_numbers_by_position`, `remove_numbers`, `keep_numbers_by_position`, `keep_random_numbers`

Manual p. 223

remove_extra_spaces

table `table.remove_extra_spaces()`

Remove spaces at the start and end of the text, and double spaces inside it.

e.g. `table.remove_extra_spaces("name")` (marks)

Options `column=`

Other tools `str_trim`, `strip`, `trimws`

See also `make_uppercase`, `replace_values`, `make_lowercase`, `round_to`, `make_dates`

Manual p. 167; lesson 11 (p. 57)

remove_numbers

nums `nums.remove_numbers()`

Remove the numbers that match one condition (the opposite of `keep_numbers`).

e.g. `nums.remove_numbers(below=0)`

Options `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=`, `not_equal_to=`, `between=`, `one_of=` and 1 more

See also `remove_duplicate_numbers`, `remove_empty_numbers`, `remove_numbers_by_position`, `keep_numbers`, `keep_first_numbers`

Manual p. 221

remove_numbers_by_position

nums `nums.remove_numbers_by_position()`

Remove numbers by their position, counting from 1: a run, every n-th number, or a list.

e.g. `nums.remove_numbers_by_position("last")`

Options `positions=`, `to=`, `step=`

See also `remove_empty_numbers`, `remove_duplicate_numbers`, `remove_numbers`, `keep_numbers_by_position`, `replace_numbers`

Manual p. 224

remove_rows

table `table.remove_rows()`

Remove the rows that match a condition (the opposite of `keep_rows`).

e.g. `table.remove_rows("subject", equal_to="English")` (marks)

Options `column=`, `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=`, `not_equal_to=`, `between=` and 7 more

See also `remove_rows_by_position`, `remove_duplicate_rows`, `remove_rows_with_outliers`, `keep_rows`, `keep_first_rows`

Manual p. 158; lesson 10 (p. 52)

remove_rows_by_position

table `table.remove_rows_by_position()`

Remove rows by their position, counting from 1: a run, every n-th row, or a list.

e.g. `table.remove_rows_by_position([12, 40])` (marks)

Options `positions=`, `to=`, `step=`

Other tools `drop_index`

See also `remove_duplicate_rows`, `remove_rows`, `remove_rows_with_outliers`, `keep_rows_by_position`, `reverse_rows`

Manual p. 160; lesson 10 (p. 52)

grid `grid.remove_rows_by_position()`

Remove rows of the grid by their position, counting from 1.

e.g. `grid.remove_rows_by_position(1)`

Options `positions=`, `to=`, `step=`

See also `remove_columns_by_position`, `keep_columns_by_position`, `keep_rows_by_position`, `replace_numbers`, `numbers`

Manual p. 233

remove_rows_with_empty_cells

table `table.remove_rows_with_empty_cells()`

Remove rows with an empty cell (in any column, or only in the column you name).

e.g. `table.remove_rows_with_empty_cells("sales")` (shop)

Options `column=`

Other tools `drop_nulls`, `dropna`, `na_omit`

See also `count_empty_cells`, `fill_empty_cells`, `add_is_empty_column`

Manual p. 163; lesson 11 (p. 57), lesson 15 (p. 81)

remove_rows_with_outliers

table `table.remove_rows_with_outliers()`

Remove the rows whose value in a column is an outlier, and say how many and why.

e.g. `table.remove_rows_with_outliers("marks", for_each="subject")` (marks)

Options `column=`, `for_each=`, `rule=`

See also `remove_duplicate_rows`, `remove_rows_by_position`, `remove_rows`, `show_rows_with_outliers`, `keep_smallest_rows`

Manual p. 162; lesson 14 (p. 76)

rename_column

table `table.rename_column()`

Give a column a new name.

e.g. `table.rename_column("marks", to="score")` (marks)

Options `old=`, `to=`

Other tools `rename`

See also `remove_columns_by_position`, `keep_columns_by_position`, `remove_columns`, `keep_columns`

Manual p. 157; lesson 10 (p. 52)

replace_numbers

nums `nums.replace_numbers()`

Replace a number, or every number that meets one condition; `with_value=None` makes them empty.

e.g. `nums.replace_numbers(-999, with_value=None)`

Options `old=`, `with_value=`, `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=`, `not_equal_to=` and 3 more

See also `remove_numbers_by_position`, `fill_empty_numbers`, `keep_numbers_by_position`, `sort`, `remove_empty_numbers`

Manual p. 224

grid `grid.replace_numbers()`

Replace a number, or every number that meets one condition; the grid keeps its rows and columns.

e.g. `grid.replace_numbers(below=4, with_value=0)`

Options `old=`, `with_value=`, `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=`, `not_equal_to=` and 3 more

See also `remove_columns_by_position`, `swap_rows_and_columns`, `remove_rows_by_position`, `keep_columns_by_position`, `keep_rows_by_position`

Manual p. 234

replace_text

table `table.replace_text()`

Replace a piece of text inside the cells of a text column, wherever it appears.

e.g. `table.replace_text("shop", " shop", with_value="")` (fruit)

Options `column=`, `old=`, `with_value=`, `pattern=`

See also `split_column`, `add_text_part_column`, `add_date_part_column`

Manual p. 174; lesson 11 (p. 57)

replace_values

table `table.replace_values()`

Replace a value (or a list of values) in a column; `with_value=None` makes those cells empty.

e.g. `table.replace_values("subject", "Maths", with_value="Math")` (marks)

Options `column=`, `old=`, `with_value=`, `above=`, `below=`, `at_least=`, `at_most=`, `equal_to=` and 8 more

Other tools `clip`, `mask`, `recode`, `replace`

See also `remove_extra_spaces`, `round_to`, `make_uppercase`, `change_column`, `make_lowercase`

Manual p. 167; lesson 11 (p. 57)

reverse

nums `nums.reverse()`

Turn the order round: the last number comes first.

e.g. `nums.reverse()`

Options none

See also `sort`, `fill_empty_numbers`, `replace_numbers`, `remove_numbers_by_position`, `keep_numbers_by_position`

Manual p. 225

reverse_rows

table `table.reverse_rows()`

Turn the table upside down: the last row comes first.

e.g. `table.reverse_rows()` (marks)

Options none

Other tools `flip`, `flipud`, `reverse`

See also `remove_rows_by_position`, `keep_random_rows`, `keep_rows_by_position`, `remove_duplicate_rows`, `keep_last_rows`

Manual p. 160; lesson 10 (p. 52)

rotate_labels_x

chart `chart.rotate_labels_x()`

Turn the names under the x-axis so long ones do not bump into each other.

e.g. `chart.rotate_labels_x()` (marks)

Options `angle=`

See also `rotate_labels_y`, `ticks_y`, `ticks_x`, `tick_format_x`, `log_scale_y`

Manual p. 138; lesson 2 (p. 13)

rotate_labels_y

chart `chart.rotate_labels_y()`

Turn the names beside the y-axis.

e.g. `chart.rotate_labels_y(30)` (fruit)

Options `angle=`

See also `rotate_labels_x`, `tick_format_x`, `ticks_y`, `tick_format_y`, `ticks_x`

Manual p. 138

round_to

table `table.round_to()`

Round numbers to this many decimal places: in one column, a list of columns, or every number column.

e.g. `table.round_to("hours", 0)` (marks)

Options `column=`, `decimals=`

Other tools `round`

See also `replace_values`, `change_column`, `remove_extra_spaces`, `make_uppercase`, `make_lowercase`

Manual p. 168; lesson 11 (p. 57)

nums `nums.round_to()`

Round every number to this many decimal places.

e.g. `nums.round_to(2)`

Options `decimals=`

See also `absolute_value`, `log`, `square_root`, `exp`, `power`

Manual p. 219

row

table `table.row()`

One row as a plain dict of column: value, counting from 1.

e.g. `table.row(3)` (marks)

Options `position=`

See also `column`, `cell_at`, `describe_columns`, `column_names`, `show`

Manual p. 154

grid `grid.row()`

One row of the grid as new numbers, counting from 1: `row(1)` is the top row.

e.g. `grid.row(1)`

Options `position=`

See also `number_at`, `column`, `show_last_rows`, `numbers`, `show_first_rows`

Manual p. 231

running_total

nums `nums.running_total()`

Each number becomes the total so far (the cumulative sum): 1, 2, 3 becomes 1, 3, 6.

e.g. `nums.running_total()`

Options none

See also `exp`, `log`, `round_to`, `absolute_value`, `square_root`

Manual p. 220

S

save

table `table.save()`

Save the table; the end of the file name decides the type.

e.g. `table.save("clean.xlsx")` (marks)

Options `file_name`

Other tools `to_csv`, `to_excel`, `to_parquet`, `write_csv`, `write_parquet`, `write_xlsx`

See also `start_again`, `chart`, `steps`, `as_pandas`, `undo`

Manual p. 212; lesson 10 (p. 52), lesson 15 (p. 81)

chart `chart.save()`

Save the chart.

e.g. `chart.save("marks.png")` (marks)

Options `file_name`, `quality=`, `resolution=`, `dpi=`

Other tools `ggsave`, `savefig`

See also `show`, `example`, `chart`

Manual p. 112; lesson 1 (p. 10)

nums `nums.save()`

Save the numbers; the end of the file name decides the type.

e.g. `nums.save("nums.csv")`

Options `file_name`

See also `chart`, `as_numpy`, `table`, `start_again`, `steps`

Manual p. 236

scatter *see* `dots`

second_y

chart `chart.second_y()`

A second y-axis on the right, for a column measured in other units.

e.g. `chart.second_y("rain", drawn_as="bars", label="Rain (mm)")` (weather)

Options `y`, `drawn_as=`, `label=`, `colour=`

Other tools `twinx`

See also `tick_format_y`, `flip_x`, `tick_format_x`, `flip_y`, `rotate_labels_y`

Manual p. 139; lesson 8 (p. 41)

shade_between_x

chart `chart.shade_between_x()`

Shade the band between two x values, behind the data.

e.g. `chart.shade_between_x(6, 8)` (weather)

Options `low`, `high`, `colour=`

Other tools `axvspan`

See also `shade_between_y`, `line_with_slope`, `line_at_y`, `line_at_x`

Manual p. 131

shade_between_y

chart `chart.shade_between_y()`

Shade the band between two y values, behind the data.

e.g. `chart.shade_between_y(40, 60)` (marks)

Options `low`, `high`, `colour=`

Other tools `axhspan`

See also `shade_between_x`, `line_with_slope`, `line_at_y`, `line_at_x`

Manual p. 131; lesson 6 (p. 31)

shape

grid `grid.shape()`

How many rows and columns, as (rows, columns).

e.g. `grid.shape()`

Options none

See also `count_columns`, `show_first_rows`, `count_rows`, `show_last_rows`, `number_grid`

Manual p. 230

show

table `table.show()`

Show the table, every row and every column (or only the first `rows=`).

e.g. `table.show()` (marks)

Options `rows=`

See also `show_last_rows`, `show_first_rows`, `column_names`, `describe_columns`, `table`

Manual p. 152; lesson 9 (p. 48)

chart `chart.show()`

Show the chart in a window.

e.g. `chart.show()` (marks)

Options none

See also `example`, `save`, `chart`

Manual p. 112; lesson 1 (p. 10)

option show=

What each bar or point on a line stands for, when a group has many rows.

e.g. `chart.bars(x="class", y="marks", show="median")` (marks)

See also error, error_bars, fit_line, fit_curve, fit_formula

Manual p. 121; lesson 5 (p. 26)

nums `nums.show()`

Show every number.

e.g. `nums.show()`

Options none

See also show_first_numbers, show_last_numbers, random_numbers, numbers, number_at

Manual p. 215

show_first_numbers

nums `nums.show_first_numbers()`

Show the first numbers.

e.g. `nums.show_first_numbers(5)`

Options how_many=

See also show, show_last_numbers, random_numbers, number_at, numbers

Manual p. 215

show_first_rows

table `table.show_first_rows()`

Show the first rows, numbered from 1.

e.g. `table.show_first_rows(5)` (marks)

Options rows=

Other tools head

See also show_last_rows, show, table, column_names, describe_columns

Manual p. 152; lesson 9 (p. 48), lesson 16 (p. 86)

grid `grid.show_first_rows()`

Show the first rows.

e.g. `grid.show_first_rows(5)`

Options rows=

See also show_last_rows, shape, count_columns, number_at, count_rows

Manual p. 230

show_grid

chart `chart.show_grid()`

Grid lines across the chart in both directions.

e.g. `chart.show_grid()` (marks)

Options none

Other tools grid

See also show_legend, size, hide_grid, style, legend_top

Manual p. 143

show_last_numbers

nums `nums.show_last_numbers()`

Show the last numbers.

e.g. `nums.show_last_numbers(5)`

Options how_many=

See also show_first_numbers, show, number_at, count, random_numbers

Manual p. 216

show_last_rows

table `table.show_last_rows()`

Show the last rows, with their row numbers in the whole table.

e.g. `table.show_last_rows(5)` (marks)

Options rows=

Other tools tail

See also show_first_rows, show, table, column_names, describe_columns

Manual p. 152; lesson 9 (p. 48)

grid `grid.show_last_rows()`

Show the last rows.

e.g. `grid.show_last_rows(5)`

Options rows=

See also show_first_rows, number_at, shape, row, count_columns

Manual p. 230

show_legend

chart `chart.show_legend()`

The legend in its usual place, for example after `hide_legend()`.

e.g. `chart.show_legend()` (weather)

Options none

See also show_grid, hide_legend, label_legend, legend_right, label_colour_bar

Manual p. 145

show_rows_with_outliers

table `table.show_rows_with_outliers()`

Show the rows whose value in a column is an outlier, and say why (the table does not change).

e.g. `table.show_rows_with_outliers("marks", for_each="subject")` (marks)

Options column=, for_each=, rule=

See also keep_smallest_rows, remove_rows_with_outliers, keep_biggest_rows, remove_duplicate_rows, keep_random_rows

Manual p. 162; lesson 14 (p. 76)

side_by_side

sdata `sdata.side_by_side()`

Charts next to each other.

e.g. `sdata.side_by_side(a, b)` (marks)

Options charts..., same_x=, same_y=, letters=

See also one_chart_per, one_above_other, grid, zoom_box, animate

Manual p. 148; lesson 7 (p. 36), lesson 12 (p. 64)

side_histograms

chart `chart.side_histograms()`

Small histograms along the top and right edges of a dots chart.

e.g. `chart.side_histograms()` (marks)

Options none

Other tools `jointplot`

See also `all_dots`, `qq_dots`, `count_bars`, `cumulative_share`, `density`

Manual p. 127

size

chart `chart.size()`

The size of the chart.

e.g. `chart.size(width=17, height=12, unit="cm")` (weather)

Options `width=`, `height=`, `unit=`

Other tools `figsize`, `set_size_inches`

See also `style`, `show_grid`, `colour_scale`, `hide_grid`, `colours`

Manual p. 142; lesson 8 (p. 41)

smallest

nums `nums.smallest()`

The smallest number; empty values are skipped.

e.g. `nums.smallest()`

Options none

See also `total`, `biggest`, `median`, `standard_deviation`, `average`

Manual p. 227

smallest_of

table `table.smallest_of()`

The smallest number in a column; empty cells are skipped.

e.g. `table.smallest_of("temperature", for_each="city")` (weather)

Options `column`, `for_each=`, `every=`, `across=`

Other tools `min`, `minimum`

See also `total_of`, `biggest_of`, `median_of`, `standard_deviation_of`, `average_of`

Manual p. 178

smooth_line

chart `chart.smooth_line()`

A smooth line that follows the dots (LOWESS).

e.g. `chart.smooth_line()` (weather)

Options `colour=`, `thickness=`, `method=`, `smoothness=`

Other tools `loess`, `lowess`

See also `curve`, `density`, `fit_distribution`, `count_bars`, `fit_formula`

Manual p. 125; lesson 5 (p. 26)

solve

sdata `sdata.solve()`

Find the values that make an equation true.

e.g. `sdata.solve("x**3 - 2*x = 5", between=[0, 3])`

Options `equations=`, `between=`, `start=`, `method=`, `known=`, `fast=`

Other tools `bisect`, `brenth`, `brentq`, `fsolve`, `fzero`, `newton`, `nleqslv`, `polyroot`, `root`, `root_scalar`, `roots`, `uniroot`, `zeros_of`

See also `find_biggest`, `find_smallest`

Manual p. 191; lesson 19 (p. 99)

sort

nums `nums.sort()`

Put the numbers in order, smallest first; empty values go last.

e.g. `nums.sort(biggest_first=True)`

Options `biggest_first=`

See also `fill_empty_numbers`, `reverse`, `replace_numbers`, `remove_numbers_by_position`, `keep_numbers_by_position`

Manual p. 225

sort_by

table `table.sort_by()`

Put the rows in order of one or more columns; empty cells go last.

e.g. `table.sort_by("marks", biggest_first=True)` (marks)

Options `columns...`, `biggest_first=`

Other tools `arrange`, `flip`, `flipud`, `order`, `reverse`, `sort`, `sort_values`

See also `add_column`, `add_group_column`, `add_rank_column`

Manual p. 169; lesson 10 (p. 52)

split_column

table `table.split_column()`

Split a text column into several columns at a separator, such as a comma or a space.

e.g. `table.split_column("shop", separator=" ", into=["shop_kind", "shop_word"])` (fruit)

Options `column=`, `separator=`, `into=`

See also `add_text_part_column`, `replace_text`, `add_date_part_column`

Manual p. 173; lesson 11 (p. 57)

square_root

nums `nums.square_root()`

The square root of every number: 9 becomes 3.

e.g. `nums.square_root()`

Options none

See also `power`, `absolute_value`, `divide_by`, `round_to`, `times`

Manual p. 219

standard_deviation

nums `nums.standard_deviation()`

How spread out the numbers are: the sample standard deviation (divide by n - 1).

e.g. `nums.standard_deviation()`

Options none

See also `biggest`, `percentile`, `smallest`, `count_each`, `total`

Manual p. 227; lesson 13 (p. 72)

dist `dist.standard_deviation()`

The standard deviation of the distribution.

e.g. `dist.standard_deviation()`

Options none

See also `median`, `variance`, `average`, `summary`, `middle_range`

Manual p. 198

standard_deviation_of

table `table.standard_deviation_of()`

How spread out the numbers are: the sample standard deviation (divide by n - 1).

e.g. `table.standard_deviation_of("marks", for_each="class")` (marks)

Options `column`, `for_each=`, `every=`, `across=`

Other tools `sd`, `std`

See also `biggest_of`, `summary_of`, `smallest_of`, `percent_of`, `total_of`

Manual p. 179

start_30_days

sdata `sdata.start_30_days()`

Start 30 days of simplydata: an hour a day of warm-up questions, lessons and tasks.

e.g. `sdata.start_30_days()`

Options `track=`, `start_again=`

Other tools `course`, `tutorial`

See also `today`, `day`, `answers`, `next_day`

Manual p. 239; lesson 1 (p. 10)

start_again

table `table.start_again()`

Go back to the table as it was at the start.

e.g. `table.start_again()` (marks)

Options none

See also `steps`, `save`, `undo`, `chart`, `as_pandas`

Manual p. 211; lesson 10 (p. 52)

nums `nums.start_again()`

Go back to the numbers as they were at the start.

e.g. `nums.start_again()`

Options none

See also `steps`, `table`, `undo`, `chart`, `save`

Manual p. 235

steps

table `table.steps()`

List every step you did to this table, in order.

e.g. `table.steps()` (marks)

Options none

See also `undo`, `start_again`, `save`, `chart`, `as_pandas`

Manual p. 211; lesson 10 (p. 52), lesson 15 (p. 81)

nums `nums.steps()`

List every step you did, in order.

e.g. `nums.steps()`

Options none

See also `undo`, `start_again`, `table`, `chart`, `save`

Manual p. 235; lesson 13 (p. 72)

style

chart `chart.style()`

The overall look.

e.g. `chart.style("school")` (marks)

Options `name`

See also `colour_scale`, `size`, `colours`, `show_grid`, `colour`

Manual p. 142; lesson 3 (p. 16), lesson 8 (p. 41), lesson 15 (p. 81)

subtract

nums `nums.subtract()`

Take a number away from every number.

e.g. `nums.subtract(2)`

Options `number=`

See also `add`, `times`, `divide_by`, `power`, `square_root`

Manual p. 217

summary

dist `dist.summary()`

A one-row table: average, median, standard deviation, variance, skewness, excess kurtosis, entropy and the lowest and highest possible values.

e.g. `dist.summary()`

Options none

See also `variance`, `random_numbers`, `standard_deviation`, `chart`, `median`

Manual p. 198

summary_of

table `table.summary_of()`

Several measures of one or more columns at once, in one tidy table.

e.g. `table.summary_of("marks", for_each="class", show=["average", "standard deviation", "count"])` (marks)

Options `column=`, `show=`, `for_each=`, `every=`

Other tools `summarise`, `summarize`, `summary`

See also `standard_deviation_of`, `percent_of`, `biggest_of`, `count_different_values`, `smallest_of`

Manual p. 179; lesson 12 (p. 64)

surface

chart `chart.surface()`

A 3D landscape: `z=` is how high the ground is.

e.g. `chart.surface(x="east", y="north", z="height")` (island)

Options `x=`, `y=`, `z=`, `transparency=`

Other tools `plot_surface`

See also `contour`, `dots_3d`, `polar`, `band`, `pie`

Manual p. 119

survival_curves

chart `chart.survival_curves()`

The share still without the event over time (Kaplan-Meier curves).

e.g. `chart.survival_curves(time="days", event="died", colour="treatment")` (trial)

Options `time=`, `event=`, `colour=`, `confidence_band=`, `censor_marks=`, `at_risk_table=`, `transparency=`, `thickness=`

See also `survival_over_time`, `explain_survival`, `compare_survival`

Manual p. 207; lesson 18 (p. 95)

survival_over_time

table `table.survival_over_time()`

The share still without the event over time (Kaplan-Meier), with the median time.

e.g. `table.survival_over_time(time="days", event="died", for_each="treatment")` (trial)

Options `time=`, `event=`, `for_each=`

Other tools `kaplan_meier`, `survfit`

See also `survival_curves`, `compare_survival`, `explain_survival`

Manual p. 205; lesson 18 (p. 95)

swap_rows_and_columns

grid `grid.swap_rows_and_columns()`

Turn rows into columns and columns into rows (the transpose in maths).

e.g. `grid.swap_rows_and_columns()`

Options none

See also `replace_numbers`, `remove_columns_by_position`, `remove_rows_by_position`, `keep_columns_by_position`, `keep_rows_by_position`

Manual p. 234; lesson 13 (p. 72)

T

table

sdata `sdata.table()`

Start a table.

e.g. `sdata.table(sdata.example("marks"))` (marks)

Options `data=`, `sheet=`, `fast=`, `separator=`, `decimal_mark=`, `thousands_mark=`, `empty_values=`, `skip_rows=` and 2 more

Other tools `data_frame`, `DataFrame`, `read`, `read_csv`, `read_excel`, `read_json`, `read_parquet`, `read_table`, `tibble`

See also `show_first_rows`, `show_last_rows`, `show`, `column_names`, `describe_columns`

Manual p. 151; lesson 9 (p. 48), lesson 15 (p. 81)

nums `nums.table()`

Turn the numbers into a simplydata table.

e.g. `nums.table(name="marks")`

Options `name=`

See also `start_again`, `chart`, `steps`, `save`, `undo`

Manual p. 236

dist `dist.table()`

A one-row table: the distribution, its parameters and, for a fit, the standard errors, log-likelihood, AIC and BIC.

e.g. `dist.table()`

Options none

See also `chart`, `parameters`, `random_numbers`, `summary`, `variance`

Manual p. 199

tick_format_x

chart `chart.tick_format_x()`

How the numbers or dates under the x-axis are written.

e.g. `chart.tick_format_x("thousands")` (growth)

Options `style=`, `decimals=`, `out_of=`

See also `tick_format_y`, `rotate_labels_y`, `rotate_labels_x`, `second_y`, `ticks_y`

Manual p. 138

tick_format_y

chart `chart.tick_format_y()`

How the numbers beside the y-axis are written.

e.g. `chart.tick_format_y("percent")` (growth)

Options `style=`, `decimals=`, `out_of=`

See also `tick_format_x`, `second_y`, `rotate_labels_y`, `flip_x`, `rotate_labels_x`

Manual p. 139; lesson 6 (p. 31)

ticks_x

chart `chart.ticks_x()`

Where the marks on the x-axis go.

e.g. `chart.ticks_x(1, 4, 7, 10)` (weather)

Options `values...`, `labels=`, `every=`, `how_many=`, `whole_numbers=`

Other tools `set_xticks`, `xticks`

See also `ticks_y`, `log_scale_y`, `log_scale_x`, `rotate_labels_x`, `range_y`

Manual p. 137

ticks_y

chart `chart.ticks_y()`

Where the marks on the y-axis go.

e.g. `chart.ticks_y(0, 50, 100)` (marks)

Options `values...`, `labels=`, `every=`, `how_many=`, `whole_numbers=`

Other tools `set_yticks`, `yticks`

See also `ticks_x`, `rotate_labels_x`, `log_scale_y`, `rotate_labels_y`, `log_scale_x`

Manual p. 137; lesson 6 (p. 31)

times

nums `nums.times()`

Multiply every number by a number.

e.g. `nums.times(2)`

Options `number=`

See also `subtract`, `divide_by`, `add`, `power`, `square_root`

Manual p. 218

title_top

chart `chart.title_top()`

A title above the chart.

e.g. `chart.title_top("My class marks")` (marks)

Options `text`

Other tools `ggtitle`, `set_title`, `suptitle`, `title`

See also `caption_bottom`, `label_x`, `label_y`, `note`, `numbers_on_bars`

Manual p. 131; lesson 2 (p. 13)

today

sdata `sdata.today()`

Show today's day of the 30 days of simplydata: warm-up questions, what to do and a task.

e.g. `sdata.today()`

Options `follow_calendar=`

Other tools `course`, `tutorial`

See also `start_30_days`, `day`, `answers`, `next_day`

Manual p. 240; lesson 1 (p. 10)

total

nums `nums.total()`

All the numbers added up; empty values are skipped.

e.g. `nums.total()`

Options none

See also `median`, `smallest`, `average`, `biggest`, `standard_deviation`

Manual p. 226

total_of

table `table.total_of()`

All the numbers in a column added up; empty cells are skipped.

e.g. `table.total_of("count", for_each="shop")` (fruit)

Options `column`, `for_each=`, `every=`, `across=`

Other tools `sum`

See also `median_of`, `smallest_of`, `average_of`, `biggest_of`, `count_each`

Manual p. 178; lesson 12 (p. 64)

turn_columns_into_rows

table `table.turn_columns_into_rows()`

Turn some columns into rows: each value gets its own row, with its old column name beside it.

e.g. `table.turn_columns_into_rows(["temperature", "rain"])` (weather)

Options `columns=`, `names_to=`, `values_to=`

Other tools `gather`, `melt`, `pivot_longer`, `stack`, `unpivot`

See also `turn_rows_into_columns`, `add_rows_from`, `join_with`

Manual p. 210; lesson 12 (p. 64)

turn_rows_into_columns

table `table.turn_rows_into_columns()`

Turn rows into columns: each different value in `names_from=` becomes a new column, filled from `values_from=`.

e.g. `table.turn_rows_into_columns(names_from="city", values_from="temperature")` (weather)

Options `names_from=`, `values_from=`

Other tools `pivot`, `pivot_table`, `pivot_wider`, `spread`, `unstack`

See also `turn_columns_into_rows`, `add_rows_from`, `join_with`

Manual p. 210; lesson 11 (p. 57)

U

undo

table `table.undo()`

Take the last step back.

e.g. `table.undo()` (marks)

Options none

See also `steps`, `start_again`, `save`, `chart`, `as_pandas`

Manual p. 211; lesson 10 (p. 52)

nums `nums.undo()`

Take the last step back.

e.g. `nums.undo()`

Options none

See also `steps`, `start_again`, `table`, `chart`, `save`

Manual p. 235; lesson 13 (p. 72)

V

values_at

table `table.values_at()`

Estimate a column at places between the known points, by joining the dots.

e.g. `table.values_at("population", at={"year": [2001, 2003, 2011]}, for_each="country")` (growth)

Options `column=`, `at=`, `joined_by=`, `smoothness=`, `for_each=`, `outside=`

Other tools `approx`, `approxfun`, `CubicSpline`, `interp`, `interp1d`, `interpolate`, `make_interp_spline`, `pchip`, `PchipInterpolator`, `spline`, `splinefun`

See also `map_of`

Manual p. 192; lesson 20 (p. 104)

variance

dist `dist.variance()`

The variance (the standard deviation squared).

e.g. `dist.variance()`

Options none

See also `standard_deviation`, `summary`, `median`, `random_numbers`, `average`

Manual p. 198

violin *see* violins

violins

chart `chart.violins()`

Shapes that are wide where many values are.

e.g. `chart.violins(x="class", y="marks")` (marks)

Options `x=`, `y=`, `colour=`, `transparency=`, `thickness=`

Also written `violin`

Other tools `geom_violin`, `violinplot`

See also `boxes`, `heatmap`, `histogram`, `pie`, `area`

Manual p. 116; lesson 4 (p. 21)

W

words

sdata `sdata.words()`

Print every simplydata word with a short example.

e.g. `sdata.words()`

Options none

See also `help`, `lesson`

Manual p. 238; lesson 8 (p. 41)

Z

zoom_box

chart `chart.zoom_box()`

A small box showing part of the chart close up.

e.g. `chart.zoom_box(x_between=[2, 4], y_between=[40, 60])` (marks)

Options `x_between=`, `y_between=`, `where=`

See also `grid`, `animate`, `one_above_other`, `side_by_side`, `one_chart_per`

Manual p. 150; lesson 8 (p. 41)

Coming from another tool?

Names from pandas, Polars, R, NumPy, SciPy, matplotlib and seaborn, and everyday maths words. In simplydata they never run: `sdata.help("groupby")` (or typing `table.groupby()`) points to the simplydata word. Here each name points to the word and the manual page that explains it. A name that is also a simplydata word is written with its object in front, such as `table.table`.

A

abline → `chart.line_with_slope()`, p. 130
adonis → `table.compare_profiles()`, p. 205
adonis2 → `table.compare_profiles()`, p. 205
agg → `table.average_of()`, p. 177
aggregate → `table.average_of()`, p. 177
anderson → `table.compare_shape_to()`, p. 185
annotate → `chart.note()`, p. 133
anova → `table.compare_averages()`, p. 183
aov → `table.compare_averages()`, p. 183
append → `table.add_rows_from()`, p. 209
approx → `table.values_at()`, p. 192; `table.fill_empty_cells()`, p. 164
approxfun → `table.values_at()`, p. 192; `table.fill_empty_cells()`, p. 164
arange → `sdata.numbers()`, p. 214
argmax → `sdata.find_biggest()`, p. 191
argmin → `sdata.find_smallest()`, p. 190
arrange → `table.sort_by()`, p. 169
array → `sdata.numbers()`, p. 214
as_date → `table.make_dates()`, p. 166
as_numeric → `table.make_numbers()`, p. 165
assign → `table.add_column()`, p. 170
astype → `table.make_numbers()`, p. 165
at → `table.cell_at()`, p. 154
axhline → `chart.line_at_y()`, p. 130
axhspan → `chart.shade_between_y()`, p. 131
axline → `chart.line_with_slope()`, p. 130
axvline → `chart.line_at_x()`, p. 129
axvspan → `chart.shade_between_x()`, p. 131

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barh → `chart.bars()`, p. 113
barplot → `chart.bars()`, p. 113
bind_rows → `table.add_rows_from()`, p. 209
bisect → `sdata.solve()`, p. 191
boxplot → `chart.boxes()`, p. 116
brenth → `sdata.solve()`, p. 191
brentq → `sdata.solve()`, p. 191

C

cast → `table.make_numbers()`, p. 165
cdf → `dist.chance()`, p. 195
chi2_contingency → `table.compare_counts()`, p. 185
chisq_test → `table.compare_counts()`, p. 185
choice → `nums.keep_random_numbers()`, p. 222
clip → `table.replace_values()`, p. 167
cmap → `chart.colour_scale()`, p. 141
colnames → `table.column_names()`, p. 153
colormap → `chart.colour_scale()`, p. 141
columns → `table.column_names()`, p. 153
concat → `table.add_rows_from()`, p. 209
contourf → `chart.contour()`, p. 118
cor → `table.correlation_between()`, p. 182
cor_test → `table.correlation_between()`, p. 182
corr → `table.correlation_between()`, p. 182
table.count → `table.count_rows()`, p. 176
countplot → `chart.count_bars()`, p. 126
course → `sdata.start_30_days()`, p. 239; `sdata.today()`, p. 240
cox → `table.explain_survival()`, p. 206
coxph → `table.explain_survival()`, p. 206
crosstab → `table.compare_counts()`, p. 185
CubicSpline → `table.values_at()`, p. 192; `table.fill_empty_cells()`, p. 164
curve_fit → `table.fit_formula()`, p. 189; `chart.fit_formula()`, p. 123
cut → `table.add_group_column()`, p. 171

D

data_frame → `sdata.table()`, p. 151
DataFrame → `sdata.table()`, p. 151
dbinom → `dist.density_at()`, p. 196
describe → `table.describe_columns()`, p. 153
diff → `table.add_change_column()`, p. 175
distinct → `table.remove_duplicate_rows()`, p. 161
dnorm → `dist.density_at()`, p. 196
drop → `table.remove_columns()`, p. 155
drop_duplicates → `table.remove_duplicate_rows()`, p. 161
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drop_nulls → `table.remove_rows_with_empty_cells()`, p. 163
dropna → `table.remove_rows_with_empty_cells()`, p. 163
dtypes → `table.describe_columns()`, p. 153

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ecdfplot → chart.cumulative_share(), p. 128
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fdr → table.add_adjusted_p_value_column(), p. 186
figsize → chart.size(), p. 142
figure → sdata.chart(), p. 111; chart.lines(), p. 113
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fill_nan → table.fill_empty_cells(), p. 164
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geom_hline → chart.line_at_y(), p. 130
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qcut → table.add_group_column(), p. 171
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Old names

Words and options that were renamed. The old name still works for one release, with a note that gives the new name.

all_points → chart.all_dots(), p. 127
band= → confidence_band= in chart.fit_line(), p. 122
between= → for_each= in table.compare_averages(), p. 183; table.compare_each_pair(), p. 183; table.compare_counts(), p. 185
big_data= → fast= in sdata.chart(), p. 111; sdata.table(), p. 151
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count_empty → table.count_empty_cells(), p. 163; nums.count_empty_numbers(), p. 217
done → table.as_pandas(), p. 212; nums.as_numpy(), p. 237
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remove_empty_rows → table.remove_rows_with_empty_cells(), p. 163
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show_first → table.show_first_rows(), p. 152; nums.show_first_numbers(), p. 215; grid.show_first_rows(), p. 230
show_last → table.show_last_rows(), p. 152; nums.show_last_numbers(), p. 216; grid.show_last_rows(), p. 230